Wh-movement paths and oblique extraction in Mam (Mayan)

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Abstract

This paper examines the Ā-extraction of oblique arguments and adjuncts in San Juan Ostuncalco Mam (SJO Mam; Mayan), focusing on the enclitic *=(y)a'*, which (optionally) appears on predicates when certain obliques are extracted. We describe the distribution of *=(y)a'* in monoclausal and long-distance extraction and show that, with the exception of temporals, the extraction of the oblique arguments and adjuncts licenses the enclitic, which may occur multiple times along a movement dependency, including within a single clause. We argue that *=(y)a'* provides a morphological footprint of the path taken by an extracted oblique. Specifically, we analyze *=(y)a'* as the spellout of Ā-agreement between the extracted oblique and intermediate heads along the clausal spine, providing evidence that Ā-movement proceeds successive-cyclically through the extended verbal domain, both within and across clauses.

Keywords: *wh*-agreement, Ā-movement, obliques, successive cyclicity, Mayan, Mam

Word count: 11,211

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1 Introduction

1.1 Extraction morphology and movement paths

Pronounced (sub)copies of Ā-moved phrases and other overt morphosyntactic reflexes of extraction are a common source of evidence used to diagnose *successive-cyclic* movement (Chomsky, 1973, 1977) in local and long-distance Ā-dependencies. Two illustrative examples of this kind of extraction reflex come from the Bor dialect of the Nilotic language Dinka in (1) and the Mayan language K'iche' in (2). In Dinka, the third-person plural pronoun *ké* occurs when plural nominals are extracted; in K'iche', the *fronting particle* *wi* occurs when certain adjuncts are extracted.

(1) Dinka (van Urk, 2018, 945)

Yè	kɔ̂ɔc-kò	cíi	Bôl	**ké**	tîiŋ
be.3SG	people-which	PRF.OV	Bol.GEN	3PL	see.NF

'Which people has Bol seen?'

(2) K'iche' (Mendes and Ranero 2021, 2, adapted from Velleman 2014, 41-12)

Jawi	x-at-b'ee	**wi**	iwiir?
where	COM-B2SG-go	**FP**	yesterday

'Where did you go yesterday?'

Although these reflexes occur in unrelated languages and differ in their morphological form and distribution, they are also strikingly similar in several respects: in addition to both *ké* and *wi* being associated with the extraction of a constituent, both morphemes surface in or around the verbal domain, and in cases of long-distance extraction, both occur in successive clauses along the movement dependency (see van Urk 2015, 2018; van Urk and Richards 2015 on Dinka; England 1997; Velleman 2014; Can Pixabaj 2015; Mendes and Ranero 2021 on K'iche'; and Georgi 2014; van Urk 2020 and references therein for further examples of similar phenomena).

Despite this surface similarity, work on these phenomena in Dinka and K'iche' has used them to support contrasting claims about the path of Ā-movement. van Urk (2018) argues that the distribution of Dinka *ké* provides evidence that Ā-movement proceeds through the verbal domain, with intermediate stops at Spec,*v*P. Mendes and Ranero (2021), by contrast, argue that the distribution of K'iche' *wi* provides evidence that the only intermediate stopover in the movement path is in the clausal domain (Spec,CP), with no intermediate stopover in the verbal domain. These two movement paths are schematized below in (3) and (4):

(3) Successive-cyclic movement via Spec,*v*P and Spec,CP (e.g., Chomsky 1986; van Urk 2018)

[$_{CP}$ WH … [$_{vP}$ <WH> … [$_{CP}$ <WH> … [$_{vP}$ <WH> … [$_{XP}$ <WH>]]]]]

(4) Successive-cyclic movement via Spec,CP only (e.g., Keine 2016; Mendes and Ranero 2021)

[$_{CP}$ WH … [$_{vP}$ … [$_{CP}$ <WH> … [$_{vP}$ … [$_{XP}$ <WH>]]]]]

Against this background, this paper examines extraction morphology in San Juan Ostuncalco Mam (SJO; Mayan; Guatemala; ISO 639-3: `mam`). SJO marks the Ā-movement of certain non-core arguments and adjuncts by means of an enclitic *=(y)a'* (England 1989). We see this in (5). In (5a), the phrase *tu'n xb'uy* 'with a rag' occurs in clause-final position, and in (5b), the *wh*-expression *alqu'n* 'with what' occurs in clause-initial position and *=(y)a'* (optionally) appears on the predicate.

(5) Oblique extraction in San Juan Ostuncalco Mam

a. *El tsu'n Li'y spik'b'il tu'n xb'uy.*

∅-∅-el	t-su-'n	Li'y	spik'b'il	[$_{\text{OBL}}$	t-u'n	xb'uy]
COMPL-B2/3SG-DIR	A2/3SG-clean-DS	María	window		A2/3SG-RN:INS	rag

'María cleaned the window with a rag'

b. *Alqu'n el tsu'n(a') Li'y spik'b'il?*

al-qu'n$_i$	∅-∅-el	t-su-'n**(=a')**	Li'y	spik'b'il	___$_i$
what-RN:INS	COMPL-B2/3SG-DIR	A2/3SG-clean-DS=MVMT	María	window	

'With what did María clean the window?'

The empirical goal of this paper is to provide a novel and detailed description of non-core argument and adjunct extraction in SJO Mam, based on new elicited data. We examine both monoclausal and long-distance extraction and identify three main descriptive generalizations: (i) with the exception of temporals, the Ā-extraction of adjuncts and non-core arguments licenses *=(y)a'*; (ii) in long-distance extraction, *=(y)a'* may appear in both the matrix and embedded clauses (depending on the the structural size of the embedded clause); and (iii) within a single clause, *=(y)a'* may appear on both the main verb and a so-called "directional" auxiliary.

The analytical goal is to provide a formal account of these patterns and, in particular, to establish what *=(y)a'* reveals about the path of extraction in SJO Mam. This question is especially interesting given that Mam and K'iche' are closely related Eastern Mayan languages, and also given the apparent similarity between SJO *=(y)a'* and K'iche' *wi*: both are verbal markers associated with the extraction of non-core/oblique arguments and adjuncts. We argue, however, that despite these similarities, *=(y)a'* patterns more like Dinka *ké* in (1) rather than with K'iche' *wi* in (2). Specifically, the distribution of *=(y)a'* provides evidence that an extracted phrase passes through intermediate positions in the extended verbal domain, as schematized in (3). More generally, the SJO facts illustrate that superficially similar extraction reflexes need not have the same derivational source: even closely related languages may employ similar morphology in systems that provide different evidence about the path of Ā-movement.

This paper is organized as follows. §2 introduces core argument, non-core argument, and adjunct extraction in the SJO Mam. §3 provides evidence that the movement enclitic *=(y)a'* in (5) marks the path of *wh*-movement, which we argue proceeds through the extended verbal projection in both local and long-distance movement. §4 analyzes *=(y)a'* as Ā-agreement between a moved oblique XP and intermediate heads in the extended verbal projection. §5 discusses implications and alternative analyses of the data discussed in this paper and situates these findings within the broader typology of oblique extraction morphology in the Eastern Mayan languages, and §6 concludes. The remainder of this section introduces SJO Mam, some analytical assumptions, and the methodology used to collect the data.

1.2 Mam background and morphosyntax

The empirical focus of this paper is Mam, an Eastern (K'ichean-Mamean) Mayan language of the Mamean branch spoken by approximately 600,000 people, primarily in the western highlands of Guatemala (Instituto Nacional de Estadística Guatemala, 2018). We focus on the underdocumented variety spoken in the municipality of San Juan Ostuncalco, Quetzaltenango, Guatemala, which belongs to the Southern Mam dialect area (England, 2017).

SJO Mam exhibits ergative-absolutive alignment, with case relations marked on the predicate (*head-marking*). In the Mayanist tradition, ergative (and possessive) person markers are referred to as *Set A*, and absolutive person markers as *Set B*. Some persons in both Sets A and B also involve a predicate-final enclitic *=(y)e'* or *=(y)a* (distinct from the enclitic *=(y)a'* tracking oblique extraction). Second and third person (singular and plural) and first person plural inclusive and exclusive are syncretic; these are distinguished by the enclitic. Paradigms for Sets A and B are given in Tables 1 and 2.

Person	**Singular**	**Plural**
1EXCL	*n- ∼ w- … =(y)e'*	*q- … =(y)e'*
1INCL		*q-*
2	*t- … =(y)a*	*k- … =(y)e'*
3	*t-*	*k-*

Table 1: Set A person marking

Person	**Singular**	**Plural**
1EXCL	*chin … =(y)e'*	*qo … =(y)e'*
1INCL		*qo*
2	*tz'- ∼ ∅- … =(y)a*	*chi … =(y)e'*
3	*tz'- ∼ ∅-*	*chi*

Table 2: Set B person marking

Like all Mam varieties, SJO has VSO(X) word order (England 1991, see also Elkins 2025): in broad-focus transitives with two overt arguments, the predicate/verb (V) precedes the subject (S), which in turn precedes the object (O); non-core arguments and adjuncts (X) follow any overt core arguments. In finite root clauses, aspect/mood (AM) and negation (NEG) markers appear clause-initially, followed by Set B agreement. A directional auxiliary (DIR) may intervene between Set B and the verbal complex (see Elkins et al. 2025 for an analysis of directionals in Mam). The verbal complex itself consists of a Set A prefix attached to the verb stem, which itself may host suffixes marking transitivity. Enclitics—including person markers and the movement enclitic *=(y)a'*—follow these suffixes. The basic clausal template is shown in (6), and a full finite transitive VSOX clause is given in (7).

(6) Mam clausal template
AM/NEG SET.B DIR $_V$[SET.A-**Stem**-suffixes]=enclitics (*subject*) (*object*)

(7) *El tsu'n Li'y spik'b'il tu'n xb'uy.*

∅-∅-el	t-su-'n	[Li'y]$_S$	[spik'b'il]$_O$	[t-u'n	xb'uy]$_X$
COMPL-B2/3SG-DIR	A2/3SG-clean-DS	María	window	A2/3SG-RN:INS	rag

'María cleaned the window with a rag.'

Non-core arguments and adjuncts in Mam are typically introduced by a *relational noun* (RN), such as *u'n* in (7) above. Relational nouns exhibit several preposition-like functions, and are additionally marked with Set A markers that agree with their complements; for example, in (7), *xbu'y* 'rag' is co-indexed by the third-person singular Set A prefix *t-* on the relational noun *u'n*.

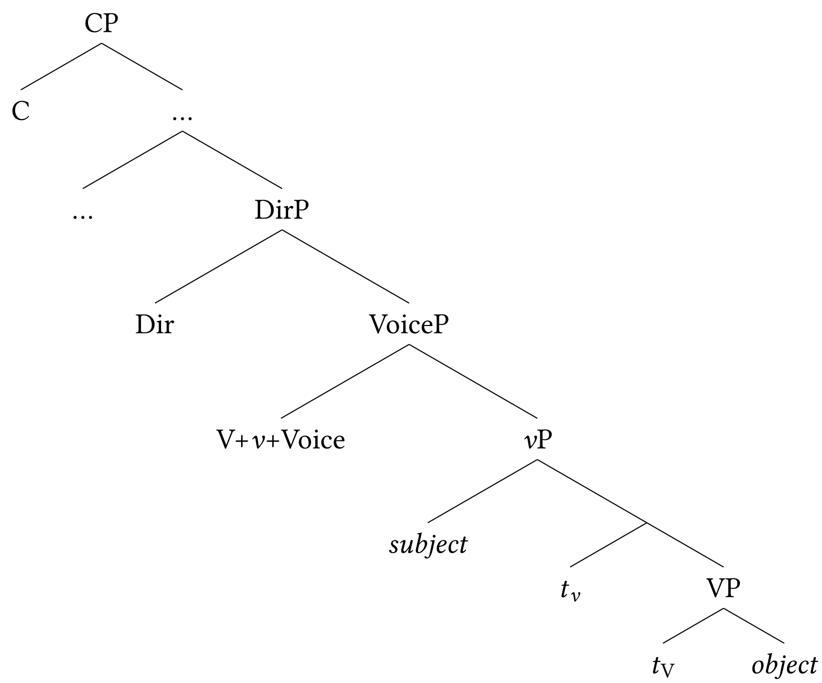
1.3 Structural assumptions

We outline here the structural assumptions adopted in this paper. While the precise hierarchy of functional structure in Mam remains an open question, we follow recent work on Mam clausal syntax (Scott

2023; Elkins 2023, 2025; Elkins et al. 2025) in assuming that verb-initiality is derived through roll-up head movement (Baker, 1985) of the verb to a peripheral position in the extended verbal domain. For concreteness, we assume a structure in which VoiceP dominates *v*P, with the external argument introduced in the specifier of *v*P.[1] The resulting VSO order is derived by head movement of the verb through *v* to Voice, placing the verb to the left of the external argument. A directional projection (DirP), headed by the directional auxiliary, dominates VoiceP, accounting for the linear positioning of directionals before the verb stem in Mam. Directionals contribute additional spatiotemporal information about the event denoted by the verb and are present across Mayan (see England 1983; Elkins et al. 2025 for discussion on directionals in Mam).

The resulting derivation of predicate-initial word order is schematized in (8).[2]

(8)



We additionally assume that oblique arguments and adjuncts, which are almost always introduced by a relational noun, may merge as either arguments or adjuncts. The relevant obliques—namely, those that trigger the movement enclitic when extracted—form a natural class in that they bear an oblique case feature [OBL]. Though there are various ways to encode this, we assume here that this feature is assigned locally by the relational noun, which assigns oblique case to its complement, as in (9).[3]

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[1]The structure in (8) differs from the standard structure where Voice introduces the external argument (e.g. Harley 2013, 2017; Burukina and Polinsky 2025). However, it parallels proposals in which a higher functional projection in the extended verbal domain dominates a *v*P layer responsible for introducing the external argument; this higher projection is labelled VoiceP in Brodkin (2022, 2025); Pietraszko (2023); Erlewine and Sommerlot (2025); Hsieh (2025), v_CP in Richards (2010), TrP (transitivity phrase) in Brown 2024a, and ssP (status suffix phrase) in, e.g., Clemens and Coon (2018); Scott (2023); Elkins (2023).

[2]Not shown here is an additional operation of *object raising*, independently motivated by syntactic ergativity/EEC-sensitivity in so-called high-ABS Mayan languages like Mam (e.g. Coon et al., 2014; Royer, 2022; Royer and Coon, 2025; Elkins, 2025; Myers, 2025). Because it does not influence the present discussion, we remain agnostic as to whether this object raising is overt and rightward, under a right-oriented specifiers approach (Aissen, 1992; Little, 2020; Scott, 2023; Elkins, 2025), or whether it is covert and leftward (Coon et al., 2014).

[3]Temporal adjuncts such as *ewi* 'yesterday', conversely, are not introduced by relational nouns. As we will see in §2.2.6, this class of temporal elements does not trigger the movement clitic when extracted.

(9) RN assigns [OBL]
$[_{\text{RNP}}$ RN^0 $\text{DP}_{[\text{OBL}]}$ $]$

1.4 A note on methodology

All uncited data in this paper come from elicitations with three SJO Mam speakers conducted between 2023 and 2026, either in situ in Quetzaltenango and San Juan Ostuncalco, Guatemala, or via telecommunications platforms. Elicitations were typically conducted in one-hour sessions and followed hypothesis-driven fieldwork methodologies (e.g., Matthewson 2004; Davis et al. 2014; Bochnak and Matthewson 2020). All consultants are bilingual speakers of Mam and Spanish, and elicitation was conducted in Spanish. Although substantial dialectal variation has been reported across Mam varieties (e.g., Cojtí and England 1986; England 1989, 2017; Simon 2019), all consultants were born and raised in San Juan Ostuncalco, minimizing potential dialectal variation.

2 Extraction in Mam

This section provides an overview of *wh*-movement (*extraction*) in SJO Mam, beginning with the extraction of core arguments in §2.1 and then turning to the extraction of non-core arguments and adjuncts in §2.2, which is the focus of this paper. A more detailed discussion of a subset of these non-core and adjunct extraction configurations follows in §3.

2.1 Core argument extraction

Wh-questions in Mam involve obligatory extraction of a *wh*-expression to clause-initial position.[4] The extraction of absolutives (intransitive subjects and transitive objects) is unmarked: apart from the clause-initial position of the extracted constituent, no additional morphosyntactic reflexes are associated with extraction, as shown in (10) and (11).

(10) Absolutive extraction: intransitive subjects

a. *Kyim jun xjaal.*
∅-∅-kyim jun xjaal
COMPL-B2/3S-die INDF man
'A man died'

b. *Alkye kyim?*
alkye ∅-∅-kyim ___
who COMPL-B2/3S-die
'Who died?'

[4] Throughout this paper, we model *wh*-question formation as a fronting process in which the *wh*-expression moves from a clause-internal argument or adjunct position to the clause periphery (Spec,CP). An alternative analysis, however, has been argued for at least some Mayan languages, where *wh*-questions are derived as pseudo-clefts: the *wh*-word takes the form of a one-place non-verbal predicate that selects a (typically headless) relative clause complement (See Royer et al. 2026 for a recent implementation). As both analyses involve Ā-movement—overt *wh*-movement in the first case and relative operator movement in the second, we believe that the empirical generalizations developed here are compatible with either derivation, and we leave the question of the source of *wh*-initiality in Mam to future work.

(11) Absolutive extraction: direct objects

a. *Tjaq' Li'y tpjel ja'.*

∅-∅-t-jaq'	Li'y	tpjel	ja'
COMPL-B2/3SG-A2/3SG-open	María	door	house

'María opened the door.'

b. *Titi' tjaq' Li'y?*

titi'	∅-∅-t-jaq'	Li'y	___
what	COMPL-B2/3SG-A2/3SG-open	María	

'What did María open?'

Ergative extraction is not straightforwardly possible. That is, Mam, like many other Mayan languages, is sensitive to the *Ergative Extraction Constraint* (EEC; see, e.g., Coon et al. 2014, 2021; Aissen 2017 and references therein). Thematic agents are instead extracted by way of a special antipassive construction. Because antipassivized predicates are formally intransitive, the thematic agent is realized as an intransitive subject (i.e., an absolutive argument), and the theme argument is "demoted" to an oblique phrase introduced by a relational noun. The verb is additionally marked with the intransitive/antipassive suffix *-n* 'ANTIP' (12).

(12) EEC effects ameliorated by antipassive construction

a. *Jaw tjq'o'n Li'y tjpel ja'.*

∅-∅-jaw	t-jq'o-'n	Li'y	tjpel	ja'
COMPL-B2/3SG-DIR	A2/3SG-open-DS	María	door	house

'María opened the door.'

b. *Alkye jq'on te tjpel ja'?*

alkye	∅-∅-jq'o-**n** ___	[$_{\text{OBL}}$	**t-e**	tjpel	ja']
who	COMPL-B2/3S-open-ANTIP		A2/3S-RN:PAT	door	house

'Who opened the door?'

Core-argument extraction in SJO therefore always targets an absolutive argument. Importantly, the movement enclitic *=(y)a'*, as seen above in (5), does *not* appear in clauses involving core-argument extraction. As we will see in more detail in §2.2, it is restricted to clauses involving the extraction of a subset of oblique/non-core arguments and adjuncts.

2.2 Non-core argument and adjunct extraction

We now turn to the extraction of non-core arguments and adjuncts (henceforth *obliques*) in SJO Mam, paying particular attention to the kinds of oblique that license the movement enclitic *=(y)a'* in (5). We show that *=(y)a'* is licensed by the extraction of instruments (§2.2.1), benefactives and datives (§2.2.2), locatives (§2.2.3), reasons and purposes (§2.2.4), and manners (§2.2.5), but not with temporals (§2.2.6).

We also note here that the movement enclitic discussed here is not restricted to SJO or Southern Mam: it is also attested in Western Mam, for example in Tacaná Mam (Munson, 1984), but is absent from Northern Mam (England, 1989). Elsewhere in the Mamean branch, we also find morphology that tracks the extraction of obliques in Ixil (Adell 2019).

2.2.1 Instruments

Instruments are introduced with the relational noun *u'n/qu'n* 'rn:ins' (13a). As shown in (13b), extraction of an instrument optionally triggers the movement enclitic *=(y)a'*.[5] The example in (13a) also shows that the movement enclitic is impossible when the instrument adjunct remains *in situ*. Thus, the enclitic is associated specifically with *extraction*, rather than with the presence of the oblique itself.

(13) Instrument extraction

a. *El tsu'n(*a') Li'y spik'b'il tu'n xb'uy.*

∅-∅-el	t-su-'n(***=a'**)	Li'y	spik'b'il	[t-u'n	xb'uy]
compl-b2/3sg-dir	a2/3sg-clean-ds=mvmt	María	window	a2/3sg-rn:ins	rag

'María cleaned the window with a rag'

b. *Alqu'n el tsu'n(a') Li'y spik'b'il?*

al-qu'n	∅-∅-el	t-su-'n(**=a'**)	Li'y	spik'b'il	___
what-rn:ins	compl-b2/3sg-dir	a2/3sg-clean-ds=mvmt	María	window	

'With what did María clean the window?'

These examples also illustrate an independent property of Mam relational nouns that we will see throughout this paper. As discussed above in §1.2, relational nouns ordinarily precede their nominal complements, as in *t-u'n xb'uy* 'with a rag' in (13a). However, when the complement is a *wh*-expression, the order is inverted, yielding *al-qu'n* 'with what' in (13b). In these inverted configurations, the Set A marker on the relational noun is also replaced by a special *q-* prefix. This pattern is an instance of so-called *pied piping with inversion* (Smith Stark, 1988) or *secondary wh-movement* (Heck, 2008), a phenomenon common in Mayan languages and Mesoamerican languages more broadly. We do not pursue an analysis of this phenomenon here, but see Heck 2008; Coon 2009; Aissen and Polian 2025; Royer et al. 2026 for different analytical approaches to pied-piping with inversion.

2.2.2 Benefactives and datives

Benefactives and datives are both introduced by the relational noun *e*; their corresponding complex *wh*-form is *alqe*. In (14), we see that the extraction of a benefactive optionally triggers the movement enclitic *=(y)a'*; in (15) we likewise see that dative extraction also optionally triggers *=(y)a'*. Just as with instruments, examples (14a) and (15a) show that in-situ benefactives and datives cannot trigger the movement enclitic.

(14) Benefactive extraction

a. *Jaw tjq'o'n(*a') Li'y tjpel ja' te Pedro.*

∅-∅-jaw	t-jq'o-'n(***=a'**)	Li'y	tjpel	ja'	[t-e	Pedro]
compl-b2/3sg-dir	a2/3sg-open-ds=mvmt	María	door	house	a2/3sg-rn:ben	Pedro

'María opened the door for Pedro'

b. *Alqe tjaq'(a')ya tjpel ja'?*

al-qe	∅-∅-t-jaq'(**=a'**)=ya	tjpel	ja'	___
who-rn:ben	compl-b2/3sg-a2/3sg-open=mvmt=2p	door	house	

'For whom did you open the door?'

[5] Whether or not the glide <y> is present is entirely phonological: it appears if the final segment of its host is a vowel or glottal stop <'>.

(15) Dative extraction

a. *Xi' kq'o'n(*a)ye' saqchb'il kye k'wal.*

∅-∅-xi'	k-q'o-'n(***=a**)=ye'	saqchb'il	[ky-e	k'wal]
COMPL-B2/3SG-DIR	A2/3SG-give-DS=MVMT=2P	toy	A2/3P-RN:DAT	child

'Y'all gave the toys to the children'

b. *Alqe' xi' kq'o'n(a)ye' saqchb'il?*

al-qe	∅-∅-xi'	k-q'o-'n(**=a'**)=ye'	saqchb'il ___
who-RN:DAT	COMPL-B2/3SG-DIR	A2/3P-give-DS=MVMT=2P	toy

'To whom did you give the toys?'

2.2.3 Locatives

The extraction of a locative oblique also allows the use of the movement enclitic *=(y)a.* In (16a), the locative expression is *toj b'e'* 'in the road' follows the VSO string and the movement enclitic is ungrammatical. Example (16b) shows that *=(y)a'* may appear on the predicate in locative questions featuring the *wh*-word *ja* 'where'.

(16) Locative extraction

a. *E' chiyon tx'yan toj b'e'.*

∅-e'	chiyo-n(***=a'**)	tx'yan	[t-oj	b'e']
COMPL-B2/3P	bark-ANTIP=MVMT	dog	A2/3S-RN:in	road

'The dogs barked in the road.'

b. *Ja e' chiyon(a') tx'yan ewi?*

ja	∅-e'	chiyo-n(**=a'**)	tx'yan ___
where	COMPL-B2/3P	bark-ANTIP-MVMT	dog

'Where did the dogs bark?'

2.2.4 Reasons and purposes

Reason and purpose obliques are introduced by the relational noun *u'n.* The *wh*-expression *tiqu'n* 'why' is morphologically complex, consisting of the *wh*-word *ti'* 'what' and the *q*-initial *wh*-allomorph of the relational noun *u'n.* Extraction of *tiqu'n* optionally triggers the movement enclitic *=(y)a'*, both when it receives a reason interpretation (17b) and when it receives a purpose interpretation (18b).

(17) Reason extraction

a. *El tsu'n Li'y spik'b'il tu'n xb'uy tu'n k'ejil*

∅-∅-el	t-su-'n	Li'y	spik'b'il	t-u'n	xb'uy	[t-u'n
COMPL-B2/3SG-DIR	A2/3SG-clean-DS	María	window	A2/3SG-RN:INS	rag	A2/3S-RN:REASON

k'ejil]
lazy

'María cleaned the window with a rag because she is lazy'

b. *Tiqu'n el tsu'n(a') Li'y spik'b'il tu'n xb'uy?*
ti'-qu'n ∅-∅-el t-su-'n**(=a')** Li'y spik'b'il t-u'n
what-RN:REASON COMPL-B2/3SG-DIR A2/3SG-clean-DS=MVMT María window A2/3SG-RN:INS
xb'uy ___
rag
'Why did María clean the window with a rag?'

(18) Purpose extraction
a. *Tjoy Xwan tx'yol toj k'ul tu'n kxi' tk'a'yin toj k'wik.*
∅-∅-t-joy Xwan tx'yol t-oj k'ul [t-u'n k-xi'
COMPL-B2/3S-A2/3S-look.for Juan mushroom A2/3S-RN:in forest A2/3S-RN:PURP A2/3P-DIR
t-k'a'y-in t-oj k'wik]
A2/3S-sell-DS A2/3S-RN:in market
'Xwan looked for mushrooms in the forest in order to sell them in the market'

b. *Tiqu'n tjoy(a') Xwan tx'yol toj k'ul?*
ti'-qu'n ∅-∅-t-joy**(=a')** Xwan tx'yol t-oj k'ul ___
what-RN:PURP COMPL-B2/3S-A2/3S-look.for=MVMT Juan mushroom A2/3S-RN:in forest
'Why did Juan look for mushrooms in the forest?'

2.2.5 Manners

The extraction of manner obliques works similarly: the *wh*-expression for 'how' in SJO is *se'n(=tu'mel)*.[6] The use of this *wh*-expression optionally triggers the use of *=(y)a'*.

(19) Manner question with *se'n=tu'mel* 'how'
a. *Kub' tpa'n Li'y lmet*
∅-∅-kub' t-pa-'n Li'y lmet
COMPL-B2/3S-DIR A2/3S-break-DS María bottle
'María broke the bottle.'

b. *Se'ntu'mel kub' tpa'n(a') Li'y lmet?*
se'n=tu'mel ∅-∅-kub' t-pa-'n**(=a')** Li'y lmet ___
how=ENCL COMPL-B2/3SG-DIR A2/3SG-break-DS=MVMT María bottle
'How did María break the bottle?'

[6]Certain *wh*-expressions in Mam appear with optional lexically-specific emphatic enclitics; for 'how', which is *se'n*, its enclitic is *=tumil*, glossed 'ENCL' above. This enclitic is also found optionally with 'where', which is *ja(=tu'mel)*. Importantly, the behavior of oblique extraction does not hinge on whether or not this enclitic is used with the *wh*-expression. For instance, consider the following example, from a text translated from Spanish to SJO Mam (*La Venganza del Pobre/Aju' tu'mel t-xel tu'n Ajyaj*):

(i) *Se'n xb'enta' tu'na tu'n tten nim tawala?*
Se'n x-∅-b'en-t=**a'** t-u'n=a t-u'n t-ten nim t-awal=a
how DIST-B2/3S-do-PASS=**MVMT** A2/3S-RN:AGT=2S A2/3S-RN:PURP A2/3S-be much A2/3S-crop=2S
'How did you make it so that you got so many crops?' (Sp: *¿Cómo lo hiciste para tener tantos cultivos?*)

2.2.6 Temporals

In contrast to the previous cases, the extraction of temporal obliques in SJO Mam patterns differently from the extraction of other obliques, in that it does *not* trigger the appearance of *=(y)a'*. SJO has two temporal *wh*-expressions: *jtoj* is used only in non-past clauses (20b), and *jtoje* is used in past clauses (21b). The extraction of these elements does *not* license the the movement enclitic (we return to this point in §5.3).

(20) a. *Kxe'l telq'a'n Xwan chmek' najchi'j.*

k-xe'-l	t-elq'a-'n	Xwan	chmek'	najchi'j
B2/3S-DIR-POT	A2/3S-steal-DS	Juan	turkey	tomorrow

'Juan will steal the turkey tomorrow.'

b. *Jtoj kxe'l telq'a'n Xwan chmek'?*

jtoj/*jtoje	k-xe'-l	t-elq'a-'n**(*=a')**	Xwan	chmek'
when.NON.PST/when.PST	B2/3S-DIR-POT	A2/3S-steal-DS=**MVMT**	Juan	turkey

'When will Juan steal the turkey?'

(21) a. *Xi' telq'a'n Xwan chmek' ewi.*

∅-∅-xi'	t-elq'a-'n	Xwan	chmek'	ewi
COMPL-B2/3S-DIR	A2/3S-steal-DS	Juan	turkey	yesterday

'Juan stole the turkey yesterday.'

b. *Jtoje xi' telq'a'n Xwan chmek'?*

jtoje/*jtoj	∅-∅-xi'	t-elq'a-'n**(*=a')**	Xwan	chmek'
when.PST/when.NON.PST	COMPL-B2/3S-DIR	A2/3S-steal-DS=**MVMT**	Juan	turkey

'When did Juan steal the turkey?'

This section has introduced extraction in SJO Mam. We showed that, among core arguments, only absolutive arguments are straightforwardly extractable, and that absolutive extraction is not associated with any additional extraction morphology. Thematic agents, by contrast, must be demoted to absolutive arguments through an antipassive construction in order to be extracted. Extraction of a subset of non-core arguments and adjuncts optionally licenses the movement enclitic *=(y)a'*: instruments, benefactives, datives, locatives, reasons/purposes, and manners license the clitic, whereas temporals do not.

Although this paper focuses on *wh*-questions, the extraction patterns described here are not specific to *wh*-movement. *Wh*-question formation belongs to a broader class of extraction constructions in Mam, including focus movement and relativization. Focused and relativized constituents likewise undergo obligatory preposing to clause-initial position and exhibit the same general morphosyntactic profile: absolutive extraction is unmarked, ergative extraction is subject to the EEC, and extraction of the relevant oblique expressions optionally licenses *=(y)a'*. The distribution of *=(y)a'* is therefore shared across these extraction constructions.

3 Extraction morphology and *wh*-paths in Mam

This section examines the distribution of the movement enclitic *=(y)a'* in more detail, and argues that it provides a morphological footprint of the path taken by an extracted oblique. We begin with two properties of its distribution that are particularly important for understanding its relationship to the movement dependency. First, *=(y)a'* may occur multiple times within a single clause when the clause contains both a main verb and a directional auxiliary (§3.1). Second, in long-distance extraction, *=(y)a'* may occur in multiple successive clauses along the movement dependency (§3.2).

The remainder of the section establishes the structural and categorial properties of *=(y)a'*. We first show that the enclitic is associated with a position below C (§3.3) and above the verb (§3.4). We then argue that *=(y)a'* is neither a (resumptive) pronoun (§3.5) nor an "extraction-facilitating" morpheme (§3.6). We conclude by bringing these observations together in §3.7, where we summarize the empirical properties that an analysis of *=(y)a'* must capture. We conclude that the enclitic is best understood as a reflex of movement through intermediate positions along the extraction path, providing evidence that oblique *wh*-movement proceeds through the extended verbal (and directional) domain in SJO.

3.1 Multiple exponence of the enclitic

Up to this point, all of the examples we have considered have contained at most one instance of the movement enclitic *=(y)a'* per clause. The enclitic may, however, be realized multiple times within a single clause: in addition to appearing on the verb stem, it may also appear on a directional auxiliary, when one is present.[7]

In (22), the locative *wh*-word *ja* is extracted, and the movement enclitic appears both on the predicate *xhchin* 'shouts' and on the preceding directional *jaw*. Both instances of the enclitic are optional and independent of one another. Consequently, the clause permits four semantically equivalent combinations: *=(y)a'* may appear on both hosts, on neither host, or on either host alone.

(22) Multiple instances of *=(y)a'*: monoclausal context (intransitive)
Ja jaw(a') xhchin(a') Luch?

ja	∅-∅-jaw**(=a')**	xhchi-n**(=a')**	Luch
where	COMPL-B2/3S-DIR=**MVMT**	shout-ANTIP=**MVMT**	Pedro

'Where did Pedro shout?'

Multiple exponence of *=(y)a'* is not restricted to a particular lexical verb, directional auxiliary, or oblique expression. Nor is it limited to intransitive clauses such as (22): the same pattern is also found in transitive clauses, as shown in (62).

(23) Multiple instances of *=(y)a'*: monoclausal context (transitive)
Se'ntumel e' xi'(ya') telq'an(a') Li'y u'j?

se'n=tumel	∅-e'	xi'**(=ya')**	t-elq'a-'n**(=a')**	Li'y	u'j?
how=ENCL	COMPL-B2/3PL	DIR=**MVMT**	A2/3S-rob-DS=**MVMT**	María	book

'How did María steal the books?'

3.2 *=(y)a'* marks the path of movement

Cross-clausal extraction provides especially clear evidence that *=(y)a'* tracks the path of movement: when an oblique is extracted from an embedded clause, *=(y)a'* occurs in each successive clause along the move-

[7] Exponence of the movement enclitic on the directional is also documented in Tacaná Mam (Munson 1984, *apud* England 1989:299). However, as Munson does not discuss multiple exponence or optionality in detail, we leave a comparison across dialects for future research.

ment dependency. Importantly, the cases discussed in this section involve extraction from full CP embedded clauses, which are characterized by (i) an overt complementizer, such as *kye* or *qa*, (ii) aspect marking, and (iii) full person agreement on the predicate. We discuss extraction from *reduced* clausal complements in §3.3 and §3.4.

Examples (24) and (25) show that extraction of an oblique or adjunct from a full CP embedded clause licenses *=(y)a’* both on the matrix predicate and on the embedded predicate. As in local extraction contexts, *=(y)a’* is also optional in long-distance extraction.

(24) Long-distance extraction from *kye* CP: multiple spellout of *=(y)a’*
Se’ntu’mel in tma’n(a’) Li’y kye tzaj wulin(a’) Xwan weye’?

se’n=tu’mel	in	∅-t-ma-’n**(=a’)**	Li’y	[$_{CP}$	kye	∅-∅-tzaj	wuli-n**(=a’)**
how=ENCL	INC	B2/3S-A2/3S-say-DS=**MVMT**	María		COMP	COM-B2/3S-DIR	insult-ANTIP=**MVMT**
Xwan	w-e=ye’	]					
Juan	A1S-RN:DAT=1S						

‘How did María say that Juan insulted me?’

Example (25) shows that *=(y)a’* may additionally surface on directionals in both the embedded and matrix clauses.

(25) Long-distance extraction from *qa* CP: multiple spellout of *=(y)a’*
Ja kub’(a’) tximana(ya’) qa jaw(a’) ktx’jo’n(a’) kamisj?

Ja	∅-∅-kub’**(=a’)**	t-xima-n=a**(=ya’)**	[$_{CP}$	qa	∅-∅-jaw**(=a’)**
where	COM-B2/3S-DIR=**MVMT**	A2/3S-think-DS=2S=**MVMT**		COMP	COM-B2/3S-DIR=**MVMT**
k-tx’jo-’n**(=a’)**	kamisj]				
A2/3P-wash-DS=**MVMT**	shirt				

‘Where did you think that they washed the shirt?’

The occurrence of *=(y)a’* in both clauses is expected if the extracted oblique passes through intermediate positions in the embedded and matrix clauses on its way to matrix Spec,CP. More generally, if *=(y)a’* is associated with movement through a particular domain, we expect it to occur only in clauses through which this movement actually occurs. This prediction is borne out in embedded questions, where the *wh*-expression moves within the embedded clause but does not undergo movement into the matrix clause. In this case, *=(y)a’* appears on the embedded predicate but is unavailable on the matrix predicate:

(26) Embedded question: *=(y)a’* restricted to embedded clause
Mixti’ b’i’n wu’ne’ se’n kub’ tbincha’n(a’) Li’y bisiklet.

mixti’	∅-b’i-’n**(*=a’)**	w-u’n=e’	[se’n	∅-∅-kub’**(=a’)**	t-b’incha-’n**(=a’)**
NEG	B2/3S-know-PASS=**MVMT**	A1S-RN:AGT=1S	how	COM-B2/3S-DIR=**MVMT**	A2/3S-repair-DS=**MVMT**
Li’y	bisiklet]				
María	bicycle				

‘I don’t know how María repaired the bicycle’

Finally, evidence from extraction islands provides further support for the claim that *=(y)a’* tracks the path of movement. If the enclitic is associated with intermediate positions in an extraction dependency, we expect it to occur only in clauses through which the extracted phrase can actually move. Consider first that *wh*-movement in SJO is subject to standard island constraints (Ross, 1967). Examples (27a)–(27b) show a relative clause modifying an object and the formation of a subject *wh*-question, respectively. Examples (27c) and (27d) show that extraction of an argument out of the relative clause is impossible, regardless of whether the relative clause contains an antipassive (27c) or an active verb (27d).

(27) a. *Xi' telq'a'n Xwan chmek' aju tzaj klq'o'n qya toj k'wik.*

∅-∅-xi'	t-elq'a-'n	Xwan	chmek'	[RC	aju	∅-∅-tzaj	k-lq'o-'n
COMPL-B2/3S-DIR	A2/3S-steal-DS	Juan	turkey		REL	COMPL-B2/3S-DIR	A2/3P-buy-DS
qya	t-oj	k'wik]					
woman	A2/3S-RN:in	market					

'Juan stole the turkey that the women bought at the market'

b. *Alkyeqe' qya lq'on te chmek'?*

alkye=qe'	qya	lq'o-n	t-e	chmek'
which=3P	woman	buy-ANTIP	A2/3S-RN:PAT	turkey

'Which women bought the turkey?'

c. **Alkyeqe' qya xi' telq'a'n Xwan chmek' aju lq'on toj k'wik?*

alkye=qe'	qya	∅-∅-xi'	t-elq'a-'n	Xwan	chmek'	[RC	aju
which=3P	woman	COMPL-B2/3S-DIR	A2/3S-steal-ANTIP	Xwan	turkey		REL
lq'on	t-oj	k'wik]					
COMPL-B2/3S-buy-ANTIP	A2/3S-RN:in	market					

Intended: 'Which women were such that Juan stole the turkey that they bought at the market?'

d. **Alkyeqe' qya xi' telq'a'n Xwan chmek' aju tzaj klq'o'n toj k'wik?*

alkye=qe'	qya	∅-∅-xi'	t-elq'a-'n	Xwan	chmek'	[RC	aju	∅-∅-tzaj
which=3P	woman	COMPL-B2/3S-DIR	A2/3S-steal-DS	Xwan	turkey		REL	COMPL-B2/3S-DIR
k-lq'o-'n	t-oj	k'wik]						
A2/3P-buy-DS	A2/3S-RN:in	market						

Intended: 'Which women were such that Juan stole the turkey that they bought at the market?'

The same restriction is observed with oblique extraction. Example (28) involves the extraction of a locative and therefore licenses *=(y)a'*. Crucially, the enclitic appears only on the matrix predicate, and the question has only a matrix-oriented interpretation: it asks where Juan stole the turkey, rather than where the women bought it. In other words, the oblique cannot be interpreted as originating *inside* the relative clause. The absence of *=(y)a'* from the embedded predicate is expected if the enclitic marks the path of an extraction dependency: the movement dependency does not extend into the relative clause, so there is no intermediate position there for *=(y)a'* to appear.

(28) *Ja xi'(ya') telq'a'n(a) Xwan chmek' aju tzaj klq'o'n qya?*

ja	∅-∅-xi'**(=ya')**	t-elq'a-'n**(=a')**	Xwan	chmek'	[RC	aju	∅-∅-tzaj
where	COMPL-B2/3S-DIR=**MVMT**	A2/3S-steal-DS=**MVMT**	Juan	turkey		REL	COMPL-B2/3S-DIR
k-lq'o-'n	qya	]					
A2/3P-buy-DS	woman						

'Where did Juan steal the turkey that the women bought?'

This conclusion is reinforced by the ungrammaticality of parallel examples where *=(y)a'* is overtly present inside the relative clause. The number and position of enclitic instances within the island have no effect on the acceptability of these configurations, as shown in (29a). Moreover, attempts to place *=(y)a'* both inside and outside the island are rejected (29b).

(29) a. **Ja xi' telq'a'n Xwan chmek' aju tzaj(a') klq'o'n(a') qya?*
*ja ∅-∅-xi' t-elq'a-'n Xwan chmek' [RC aju ∅-∅-tzaj(**=a'**)
where COMPL-B2/3S-DIR A2/3S-steal-DS Juan turkey REL COMPL-B2/3S-DIR=**MVMT**
k-lq'o-'n(**=a'**) qya]
A2/3P-buy-DS=**MVMT** woman
Intended: 'Where did Juan steal the turkey that the women bought?'

b. **Ja xi'(ya') telq'a'n(a') Xwan chmek' aju tzaj(a') klq'o'n(a') qya toj k'wik?*
*ja ∅-∅-xi'(**=ya'**) t-elq'a-'n(**=a'**) Xwan chmek' [RC aju
where COMPL-B2/3S-DIR=**MVMT** A2/3S-steal-DS=**MVMT** Juan turkey REL
∅-∅-tzaj(**=a'**) k-lq'o-'n(**=a'**) qya]
COMPL-B2/3S-DIR=**MVMT** A2/3P-buy-DS=**MVMT** woman
Intended: 'Where did Juan steal the turkey that the women bought?'

Taken together, the long-distance extraction facts support the conclusion that *=(y)a'* marks the path of *wh*-movement: the enclitic occurs in clauses through which the movement dependency passes, but not in clauses that do not form part of that dependency.[8] Having established this relationship between *=(y)a'* and the movement path, we now turn to the structural position of the heads that host the enclitic. In particular, we first ask whether *=(y)a'* is associated with the clausal or verbal domain.

3.3 The locus of *=(y)a'* is lower than CP

In addition to the full CP complements discussed in the previous section, SJO also permits *reduced* clausal complements, that is, complements that are structurally smaller than CP. These reduced clauses differ in their internal structure and, consequently, in whether they license *=(y)a'* under extraction from the embedded clause. This section focuses on so-called "aspectless" embedded clauses (following the terminology of England 2017), which we argue are VoiceP-sized reduced clauses. We show that extraction of an oblique or adjunct from an aspectless clause licenses *=(y)a'* on the embedded predicate, just as extraction from a full CP may. If aspectless clauses lack the CP layer (Scott, 2023; Elkins et al., 2025), the availability of *=(y)a'* in these clauses provides evidence that the locus of the enclitic is below C^0.

Aspectless clauses are introduced by elements such as the purposive relational noun *u'n* 'in order to, for the purpose of'.[9] These clauses lack AM morphology before the verb (hence "aspectless") and exhibit split ergativity: argument marking is always Set A, regardless of whether the argument is an agent, subject, or object (referred to as "super-extended ergativity" in England 2017; Scott 2023; Elkins et al. 2025). Following Scott (2023); Elkins et al. (2025), we analyze these clauses as structurally reduced clauses that contain VoiceP, but lack the higher functional structure associated with aspect and (high) Set B agreement. In (30), an aspectless clause is introduced by *u'n* and the intransitive subject of the passive verb *k'a'yil* 'buy' bears Set A ergative/possessive marking.

[8] The island facts also show that *=(y)a'* does not ameliorate otherwise illicit extraction out of an island, as has been argued for some cases of resumption (e.g., Ross 1967; Sells 1984; Aoun et al. 2001; Asudeh 2012; Beltrama and Xiang 2016; Ackerman et al. 2018; cf. Heestand et al. 2011). See also §3.5.

[9] Following Scott (2023), we do not analyze *u'n* as a C^0 head, as would be expected of a prototypical complementizer. Instead, we assume that this relational noun takes a nominalized VoiceP as its complement and bears A2/3S agreement with that nominalized clause.

(30) *Chu'xh tu'n tk'a'yit chmek' toj kwik.*

∅-chu'xh	t-u'n	[$_{\text{VoiceP}}$	**t-**k'a'y-it	chmek'	t-oj	k'wik	]
B2/3S-be.difficult	A2/3S-RN:PURP		A2/3S-sell-PASS	turkey	A2/3S-RN:in	market	

'It's difficult to sell turkeys in the market' (≈ '…for turkeys to be sold…')

The following examples in (31) and (32) show that long-distance dependencies involving an oblique in an aspectless embedding, *=(y)a'* may appear on both the matrix and embedded predicates.

(31) SJO extraction from aspectless embedding: multiple spellout of *=(y)a'*
Jatu'mel chu'xh(a) tu'n tk'a'yit(a') chmek'?

ja=tu'mel	∅-chu'x**(=a')**	t-u'n	[$_{\text{VoiceP}}$	t-k'a'y-it**(=a')**	chmek'	]
where=ENCL	B2/3S-be.difficult=**MVMT**	A2/3S-RN:PURP		A1S-sell-PASS=**MVMT**	turkey	

'Where is it difficult to sell turkeys?'

(32) *Ja tajb'il(a') ku'xun tu'n txu'xin(a')?*

Ja	t-ajb'il**(=a')**	ku'xun	t-u'n	[$_{\text{VoiceP}}$	t-xu'xi-n**(=a')**	]
where	A2/3S-want=**MVMT**	young.man	A2/3SRN:PURP		A2/3S-swim-AP=**MVMT**	

'Where does the young man want to swim?'

The pattern in (31)–(32) closely parallels the pattern established above in §3.2 for extraction from full CPs: when an oblique is extracted from the embedded clause, *=(y)a'* may be realized both on the matrix predicate and on the predicate of the embedded clause. (The two instances are again independently optional.) Thus, the occurrence of *=(y)a'* in the embedded clause does not depend on the presence of the higher functional structure found in a full CP. These provide important evidence that the locus of *=(y)a'* is below CP.[10]

3.4 The locus of *=(y)a'* is above VP

The second type of reduced embedding we consider are *nonfinite clauses*, which are attested in all Mam varieties (see England 1989, 292 and Scott 2023, 92-93). Nonfinite clauses are characterized by the verbal suffix *-(V)l*, the obligatory absence of directionals, and the inability to license core arguments. Nominal complements must either be introduced by a relational noun or occur with a generic interpretation, which we take to involve (pseudo-)noun incorporation (see England 1983:218–219). Given the absence of directionals, aspectual morphology, as well as their inability to license definite internal arguments (by hypothesis derived via object shift to a structurally higher position; Little 2020; Royer 2022; Elkins 2025), we follow Scott (2023, §3.4.3.1) in treating nonfinite clauses as highly reduced, VP-sized complements.

In (33) we see an example of an embedded nonfinite clause; when an oblique is extracted from an nonfinite complement clause, *=(y)a'* may appear on the matrix predicate but crucially cannot appear in the embedded clause, as shown in (34).

[10]As we will see in §5, this sets Mam apart from its Eastern Mayan neighbors K'iche' and Patzún Kaqchikel, where, in a long-distance dependency involving two clauses, the presence of the oblique *fronting particle wi* in the matrix clause is contingent on the embedded clause being a full CP; extraction from reduced clauses license *wi* in the embedded reduced clause, but not in the matrix clause (Can Pixabaj, 2015; Mendes and Ranero, 2021).

(33) *Ok kq'one' Pabl jqol spik'b'il te Li'y.*

∅-∅-ok	k-q'o-'n=e'	Pabl	$[_{VP}$	jqo-**l**	spik'b'il	t-e	Li'y	]
COM-B2/3S-DIR	A2/3P-make-DS=2S	Pablo		open-**NF**	window	A2/3P-RN:BEN	María	

'Y'all made Pablo open windows for María'

(34) *Alqe ok(a') kq'on(a')ye' Pabl ichil kye tal tx'yan?*

al-qe	∅-∅-ok**(=a')**	k-q'o-n**(=a')**=ye'	Pabl	$[_{VP}$	jqo-l(***=a'**)
who-RN:BEN	COM-B2/3S-DIR=**MVMT**	A2/3P-make-DS=**MVMT**=2P	Pablo		open-INF=MVMT

spik'b'il]
window

'For whom did y'all make Pablo open windows?'

The contrast between aspectless and nonfinite embeddings shows that *=(y)a'* may occur in a VoiceP-sized clause, but not in a VP-sized clause. This places the locus of *=(y)a'* above VP but below CP. We take this locus to be a functional head that occurs both in the extended verbal domain and in the directional domain. For expository purposes, we refer to these positions as Voice^0 and Dir^0 (but see footnote 14).

3.5 *=(y)a'* is not a pronoun

Having established general structural position of *=(y)a'*, we now turn to its categorical status. We first consider whether *=(y)a'* might be a pronominal element, specifically a resumptive pronoun indexing the left-peripheral oblique. Two properties of the enclitic argue against this analysis. First, *=(y)a'* is optional, which is unexpected if it serves as a resumptive element. Second, *=(y)a'* is not otherwise used as a pronominal element in the language.

Below we see that *=(y)a'* cannot be used anaphorically in the absence of an extracted oblique. Consider the exchange in (35a)–(35c). In (35a), the locative *toj k'wik* 'in the market' is introduced in the question. In (35b), the extracted locative *atzu* 'there' appars in clause-initial position in the response, and *=(y)a'* appears on the predicate. Crucially, however, *=(y)a'* cannot appear in contexts lacking an extracted oblique, as shown in (35c). Thus, *=(y)a'* cannot itself function as an anaphoric expression referring to a previously established locative.

(35) *=(y)a'* is not an anaphor (Adapted from Mendes and Ranero 2021 example (45))

a. *Tzaj tlq'o'n xwan q'inun toj k'wik?*

∅-∅-tzaj	t-lq'o-'n	Xwan	q'inun	t-oj	k'wik
COM-B2/3S-DIR	A2/3-buy-DS	Xwan	jocote	A2/3S-RN:in	market

'Did Juan buy the jocotes in the market?'

b. *Iku', atzu tzaj tlq'o'na'*

iku'	atzu	∅-∅-tzaj	t-lq'o-'n=**a'** ___
yes	there	COM-B2/3S-DIR	A2/3-buy-DS=**MVMT**

'Yes, there he bought them.' (Ā-movement of *atzu* 'there')

c. **Iku' tzaj tlq'ona'*

*iku'	∅-∅-tzaj	t-lq'o-'n=**a'**
yes	COM-B2/3S-DIR	A2/3-buy-DS=**MVMT**

Intended: 'Yes, there he bought them.'

The island facts discussed above in §3.2 provide further evidence against a (resumptive) pronominal analysis. As shown in (29), the presence of *=(y)a'* inside an extraction island does not have an ameliorative effect (as might be expected, see footnote 8). Taken together, these facts suggest that *=(y)a'* is not a pronominal element.

3.6 The movement enclitic is not an extraction facilitator

We next consider whether *=(y)a'* might be an "extraction-facilitating" morpheme, specifically one analogous to an *Agent Focus* (AF) marker, which is found widely across Mayan languages (see, e.g., Coon et al. 2021). In many Mayan languages, including Mam (see §2), Ā-extraction of transitive subjects/agents (ergatives) is not straightforwardly possible, and requires the use of a special construction. In languages such as Patzún Kaqchikel, for instance, extraction of an ergative argument requires the predicate to bear the AF morpheme *-o*.

(36) Patzún Kaqchikel (Mendes and Ranero, 2021, 18)

a. *Achike x-∅-u-tej nu-way?
who COMPL-B3S-A3S-eat A1S-tortilla
Intended: 'Who ate my tortilla?'

b. Achike x-∅-tj-**o** nu-way?
who COMPL-B3S-eat-**AF** A1S-tortilla
'Who ate my tortilla?'

Mendes and Ranero (2021) note a superficial parallel between the AF morpheme *-o* and the oblique extraction clitic *wi* in Kaqchikel: both are involved in the Ā-extraction of an *marked* argument (versus the comparably *unmarked* absolutive arguments). However, Mendes and Ranero reject this kind of analysis, among other reasons, because unlike *-o*, *wi* occurs alongside other valency-related morphology, including the passive. SJO Mam behaves similarly: *=(y)a'* freely occurs alongside valency morphemes such as the passive *-njtz* 'PASS' (see also 19). Crucially, morphemes such as *-njtz* are in complementary distribution with other valency-related verbal suffixes, such as intransitive *-n* and transitive *-'n* (see also Pérez Vail 2014).

(37) *=(y)a'* co-occurs with valency-related suffixes
Ja'tumel e' k'a'yinjtz(a') muqin tu'n Xwan?
ja=tu'mel e' ∅-k'a'yi-njtz(**=a'**) muqin t-u'n Xwan
where=ENCL COMPL B2/3S-sell-PASS=**MVMT** tortilla A2/3S-RN:AGT Juan
'Where were the tortillas sold by Juan?'

Secondly, although Mam is sensitive to the ERGATIVE EXTRACTION CONSTRAINT (see the discussion in §1.2), it does not have a dedicated AF morpheme. Instead, ergative extraction is facilitated by means of an antipassive construction. In contexts of long-distance Ā-extraction of an ergative argument, only the embedded clause in which the extracted argument originates as the agent must be antipassivized; the matrix clause remains active (38). This shows that the antipassive is restricted to the clause containing the extracted argument and does not extend to the matrix clause.

(38) *Alkye tmaya qa el sun te spik'b'il tu'n xb'uy?*

alkye	∅-∅-t-ma=ya	[$_{CP}$	qa	∅-∅-el	su-**n**	t-e	spik'b'il
who	COM-B2/3S-A2/3S-say=2s		COMP	COM-B2/3S-DIR	clean-**ANTIP**	A2/3S-RN:PAT	window

t-u'n	xb'uy]
A2/3S-RN:AGT	rag

'Who did you say cleaned the window with a rag?'

This behavior differs from long-distance oblique extraction, where *=(y)a'* is licensed in each clause along the movement path (see §3.2). This asymmetry, together with the ability of *=(y)a'* to co-occur with valency-related morphology, argues against analyzing it as analogous to an Agent Focus morpheme.

3.7 Local summary

The facts presented in this section establish several properties that any analysis of *=(y)a'* must capture. First, the enclitic is specifically associated with oblique Ā-movement. As shown in §3.1, it may surface on the verb and, where present, on a directional auxiliary; as shown in §3.2, it may also be realized in successive clauses along a long-distance movement dependency. Thus, *=(y)a'* provides a morphological reflex of the path taken by an extracted oblique, both in local extraction, where the *wh*-expression ultimately occupies a position in the left periphery, and in long-distance extraction, where the dependency passes through multiple clauses.

Second, the distribution of *=(y)a'* across different sizes of embedded clause (summarized in Table 3) places its locus within the extended verbal domain. As shown in §3.3, the enclitic is available in VoiceP-sized aspectless clauses, establishing that its locus is below CP. Conversely, §3.4 shows that it is unavailable in VP-sized nonfinite clauses, placing its locus above VP. Taken together, these facts place the locus of *=(y)a'* above VP but below CP. We take Voice^0 and Dir^0 to be the relevant loci.

	Matrix clause	**Embedded clause**	**Example**
Long-distance extraction from full CP	✓(+clitic)	✓	(24)
Long-distance extraction from aspectless clause	✓	✓	(31)
Long-distance extraction from nonfinite clauses	✓	✗(−clitic)	(34)

Table 3: Summary of possible spellout locations of *=(y)a'* based on clause type

Finally, we argued that *=(y)a'* should not be treated as a pronominal element or as an *Agent Focus-like* morpheme that facilitates extraction. These observations provide the empirical basis for our analysis of *=(y)a'*, which we turn to now.

4 Analyzing *=(y)a'* as a reflex of successive-cyclic movement

In this section, we argue that the movement enclitic *=(y)a'* in SJO Mam is a reflex of clause-internal successive-cyclic movement (Chomsky, 1986, 1995, 2001; Rackowski and Richards, 2005; den Dikken, 2009, 2012a,b; van Urk and Richards, 2015; Keine and Zeijlstra, 2025). Specifically, we analyze *=(y)a'* as the morphological realization of Ā-agreement (e.g., Chung and Georgopoulos, 1988; Georgopoulos, 1991; Ouhalla, 1993; Chung, 1994; Baier, 2018) between a moved oblique and successive heads along the clausal spine, namely Voice^0 and Dir^0. On this analysis, each occurrence of *=(y)a'* corresponds to a distinct step in the movement path, deriving the multiple exponence pattern described in §§3.1-3.2: the enclitic may surface multiple times because the extracted oblique undergoes successive-cyclic movement through multiple intermediate positions. We model these movement steps as the consequence of intervention-based locality, adopting a Relativized Minimality (Rizzi, 1990) approach to successive-cyclic movement (e.g., Abels, 2003; Torr, 2012; Keine and Zeijlstra, 2025; Halpert and Zeijlstra, 2026).

4.1 Minimality-based locality

We adopt a Relativized Minimality (Rizzi, 1990) approach to successive cyclicity based on Abels (2003), following the reformulation proposed by Halpert and Zeijlstra (2026). In this framework, obligatory successive cyclicity follows from feature-based locality rather than phases (Chomsky, 2000, 2001): a phrase XP moving to satisfy a feature of a higher head cannot cross an intervening head bearing the same feature. This restriction, formalized as Attract Closest in (39), with Closeness defined in (40), requires XP to move through the specifier of each intervening feature-bearing (41) head en route to its final landing site.

(39) **Attract Closest** (Abels 2003, 45, adapted from Chomsky 1995, 194, Pesetsky and Torrego 2001, 362)
If a head K attracts feature F on X, X a feature bearer of F, no constituent that bears F is closer to K than X is.

(40) **Closeness** (Abels 2003, 45 adapted from Chomsky 1995, 299, Pesetsky and Torrego 2001, 362)
Y is closer to K than X is if K c-commands Y and Y c-commands X.

(41) **Feature bearers** (Abels, 2003, 45)
A is a feature bearer of feature F iff A is a constituent and the label of A contains F.

This minimality-driven successive-cyclic movement is sketched in (42)–(43). In (42), we see a one-step derivation in which XP moves directly to its final landing site crossing an intervening head (H2) bearing the relevant feature. This movement in *one fell swoop* violates Attract Closest and is therefore ruled out. In (43), by contrast, XP first moves to the specifier of the intermediate feature-bearing head before continuing to its final landing site, thereby satisfying the locality requirement.

(42) ✗ Movement in one fell swoop
a. Step 1: Agree
$[_{HP_1}\ H1_{[\bar{A}]}\ [_{HP_2}\ H2_{[\bar{A}]}\ XP_{[\bar{A}]}\ ]\ ]$

b. Step 2: Move
$[_{HP_1}\ XP_{[\bar{A}]}\ H1_{[\bar{A}]}\ [_{HP_2}\ H2_{[\bar{A}]}\ <XP>\ ]\ ]$

(43) ✓ Successive-cyclic movement

a. Step 1: Agree
$[_{\text{HP}_1}\ \text{H1}_{[\bar{\text{A}}]}\ [_{\text{HP}_2}\ \text{H2}_{[\bar{\text{A}}]}\ \text{XP}_{[\bar{\text{A}}]}\ ]\ ]$

b. Step 2: Move
$[_{\text{HP}_1}\ \text{H1}_{[\bar{\text{A}}]}\ [_{\text{HP}_2}\ \text{XP}_{[\bar{\text{A}}]}\ \text{H2}_{[\bar{\text{A}}]}\ <\text{XP}>\ ]\ ]$

c. Step 3: Agree
$[_{\text{HP}_1}\ \text{H1}_{[\bar{\text{A}}]}\ [_{\text{HP}_2}\ \text{XP}_{[\bar{\text{A}}]}\ \text{H2}_{[\bar{\text{A}}]}\ <\text{XP}>\ ]\ ]$

d. Step 4: Move
$[_{\text{HP}_1}\ \text{XP}_{[\bar{\text{A}}]}\ \text{H1}_{[\bar{\text{A}}]}\ [_{\text{HP}_2}\ <\text{XP}>\ \text{H2}_{[\bar{\text{A}}]}\ <\text{XP}>\ ]\ ]$

Following Halpert and Zeijlstra (2026), we assume that the relevant feature bearers are not restricted to phase heads. Rather, any functional head bearing the relevant movement feature may act as an intervener, forcing movement to proceed through its specifier. Obligatory successive cyclicity therefore follows from feature-based locality, and not from phases per se. Below in §4.2, we propose that in addition to C^0, there are Ā-feature-bearing heads in the extended verbal domain (Voice^0) and the directional auxiliary domain (Dir^0).[11]

4.2 Implementing the analysis

Our proposal derives the distribution of the movement enclitic in SJO Mam as follows. We assume that *wh*-expressions are base-generated in the same clause-final positions as their non-*wh* counterparts, as in (44). In addition to an [Ā] feature, the relevant oblique expressions bear a structural [OBL] Case feature (see §1.3). Movement is ultimately triggered by C^0, which bears an [Ā] feature and attracts the closest constituent bearing an [Ā] feature to Spec,CP.[12]

(44) Pre-movement configuration
$[_{\text{CP}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \text{Dir}_{[\bar{\text{A}}]}\ [_{\text{VoiceP}}\ \text{Voice}_{[\bar{\text{A}}]}\ \ldots\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ ]\ ]\ ]$

[11] An alternative approach would derive successive cyclicity in terms of phases (Chomsky, 2000, 2001). On such an analysis, Voice^0 and Dir^0 would constitute phases, requiring an extracted oblique to move through their specifiers before continuing its movement path. This issue is particularly relevant given that Mendes and Ranero (2021), based on comparable oblique extraction data from K'ichean languages, argue that there is *no* phase within the extended verbal domain (see §5 for discussion). However, beyond the clause-internal successive-cyclicity effects discussed here, we have no independent language-internal evidence for or against the status of VoiceP or DirP as phases in SJO Mam. We additionally note here that an anonymous reviewer points out that the *=(y)a'* realized on a directional element could alternatively be understood as a C-domain reflex whose morphological realization is displaced to the directional domain. This would provide a way of maintaining a phase-based analysis in which C^0 and Voice^0 are phasal, without requiring Dir^0 to constitute an independent phase, but see example (63) in §5.2 below for an example of a local dependency featuring three instances of *=(y)a'* that might challenge this kind of approach.

[12] The *wh*-expressions *ja* 'where' and *sen(=tu'mel)* 'how' are not transparently composed of a relational noun and a *wh*-expression (compare forms such as *tiq'un* 'why', which *are* complex *wh*-expressions featuring a relational noun). Recall from §1.3 that we assume relational nouns assign an [OBL] feature to their complements. To maintain the feature-based analysis adopted here, we must therefore assume that these *wh*-expressions acquire [OBL] by some other means. One possibility is that they contain a silent relational noun; another is that [OBL] is assigned by a distinct functional head in these cases.

However, C^0 cannot Agree directly with an in-situ oblique. As discussed in §4.1, Voice^0 and, where present, Dir^0 are themselves also feature bearers of [Ā]. Assuming Attract Closest in (39), these heads intervene between C^0 and the in-situ oblique. Movement must instead proceed successive-cyclically through the specifier positions of Voice^0 and Dir^0. A crucial additional assumption is that during the derivation, the *wh*-word's [OBL] feature is copied onto the Agreeing Voice and Dir heads. C^0, by contrast, is only sensitive to [Ā] features and not Case features like [OBL]. This accounts for the absence of *=(y)a'* as a C-domain reflex of extraction, as well as the absence of *=(y)a'* in cases of argument extraction, which recall from §1.2, always involves an absolutive argument (thus not bearing an oblique feature).[13]

The derivation in (45) illustrates monoclausal extraction. Voice^0 first Agrees with the in-situ oblique *wh*-word, copying its [OBL] feature (Step 1), after which the *wh*-word moves to Spec,VoiceP (Step 2). Steps 3 and 4 instantiate the same Agree-and-Move sequence in the Directional domain. At Step 5, the *wh*-word is accessible to C^0, which Agrees with it and triggers its final movement to Spec,CP (Step 6). (If no directional is present in the derivation, then Steps 3 and 4 are skipped.)

(45) Clause-internal cyclic movement

a. Step 1: Voice^0 Agrees with oblique WH, copies [OBL] feature:

$[_{\text{CP}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \text{Dir}_{[\bar{\text{A}}]}\ [_{\text{VoiceP}}\ \text{Voice}_{[\bar{\text{A}},\ \textbf{\textsc{obl}}]}\ \ldots\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ ]\ ]\ ]$

b. Step 2: WH moves to Spec,VoiceP:

$[_{\text{CP}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \text{Dir}_{[\bar{\text{A}}]}\ [_{\text{VoiceP}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]}\ \ldots\ \langle\text{WH}\rangle\ ]\ ]\ ]$

c. Step 3: Dir^0 (if present in derivation) Agrees with oblique WH, copies [OBL] feature:

$[_{\text{CP}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \text{Dir}_{[\bar{\text{A}},\ \textbf{\textsc{obl}}]}\ [_{\text{VoiceP}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]}\ \ldots\ \langle\text{WH}\rangle\ ]\ ]\ ]$

d. Step 4: WH moves to Spec,DirP (if present in derivation):

$[_{\text{CP}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Dir}_{[\bar{\text{A}},\textsc{obl}]}\ [_{\text{VoiceP}}\ \langle\text{WH}\rangle\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]}\ \ldots\ \langle\text{WH}\rangle\ ]\ ]\ ]$

e. Step 5: C^0 Agrees with oblique WH:

$[_{\text{CP}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Dir}_{[\bar{\text{A}},\textsc{obl}]}\ [_{\text{VoiceP}}\ \langle\text{WH}\rangle\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]}\ \ldots\ \langle\text{WH}\rangle\ ]\ ]\ ]$

f. Step 6: WH moves to Spec,CP:

$[_{\text{CP}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{DirP}}\ \langle\text{WH}\rangle\ \text{Dir}_{[\bar{\text{A}},\textsc{obl}]}\ [_{\text{VoiceP}}\ \langle\text{WH}\rangle\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]}\ \ldots\ \langle\text{WH}\rangle\ ]\ ]\ ]$

We analyze *=(y)a'*, as well as its null counterpart ∅, as the realization of the feature bundle [Ā, OBL] established by Agree with Voice^0 or Dir^0.[14,15]

[13] Case-sensitive extraction morphology is independently attested in other languages. See, for example, Chung (1998); Lahne (2009); Georgi (2014) on case-sensitive *wh*-agreement in Chamorro (Austronesian, Guam and Mariana Islands).

[14] We use the labels Voice^0 and Dir^0 here, but these labels should not be taken to identify the exact syntactic heads that host the relevant features. In particular, overt Voice and Directional morphology can occur alongside *=(y)a'*, suggesting that the relevant agreement-bearing heads may instead be distinct functional heads in the verbal and directional domains. We leave the precise identity of these heads open.

[15] The optionality of successive-cyclic movement reflexes is well attested cross-linguistically. See, e.g., Torrence (2012, 1172) on optional *wh*-agreement in Wolof, Mendes and Ranero (2021) and references therein on optional extraction reflexes in K'ichean, and the discussion in Georgi (2014, §2.5.3.2). We model this optionality here as the result of two competing vocabulary items,

(46) Vocabulary insertion rule for *=(y)a’*

a. /=(y)a’/ $\leftrightarrow$ [Ā, OBL] / __ Voice^0/Dir^0

b. $\varnothing \leftrightarrow$ [Ā, OBL] / __ Voice^0/Dir^0

This analysis derives the core empirical pattern observed in local extraction configurations. Because Voice^0 and Dir^0 are feature-bearing heads, oblique extraction necessarily proceeds through their specifiers. At spellout, the [Ā, OBL] feature bundle on Voice^0 or Dir^0 is realized as *=(y)a’* (or $\varnothing$). Each Agree relation therefore provides a potential site for realization of the movement enclitic.

Let us briefly address why Dir^0 should be a stop-over point on the *wh*-extraction path, on a par with Voice^0. Research on the status and behavior of directional auxiliaries in the Mayan languages is still preliminary, and our answer is therefore necessarily tentative. We suggest that the behavior of directionals in SJO is related to their various verb-like properties. First, directionals in Mayan are transparently derived from intransitive verbs of motion (see England 1983 for Mam specifically). Second, a growing body of work has shown that directionals play an important role in argument structure (e.g., in the introduction of internal arguments in Mam; Elkins et al. 2025) and in the derivation of certain verbal constructions, including indirect arguments (Mateo Toledo 2004, 2008, 2023 for Q’anjob’al) and stative predication (Henderson et al. 2018; Elias 2019 for Chuj). The empirical facts discussed in the present paper—namely, that, whatever their precise status, directionals serve as loci for successive-cyclic movement—may in turn contribute to a fuller understanding of these elements in Mayan.[16]

4.3 Extending the analysis to long-distance movement

This section extends the analysis for clause-internal successive cyclic movement in §4.2 to successive cyclic movement in long-distance dependencies. For reasons of space, the derivations below omit intermediate movement through Spec,DirP, although such movement is possible and fully compatible with our analysis.

We first schematize oblique extraction from a full CP-sized embedding, as in (24). The first steps are shown in (47). Starting with movement internal to the embedded clause, Voice^0 first Agrees with the in-situ oblique *wh*-expression, copying its [OBL] feature (Step 1). The *wh*-expression then moves to Spec,VoiceP2 (Step 2). C^0 subsequently Agrees with the oblique (Step 3), triggering movement to Spec,CP2 (Step 4).

(47) First Steps: full CP complement

a. Step 1: Voice^0 Agrees with oblique WH, copies [OBL] feature:
$[_{\text{CP2}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ \text{Voice}_{[\bar{\text{A}},\ \textbf{OBL}]}\ \dots\ \text{WH}_{[\bar{\text{A}},\ \text{OBL}]}\ ]\ ]$

b. Step 2: WH moves to Spec,VoiceP2:
$[_{\text{CP2}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ \text{WH}_{[\bar{\text{A}},\ \text{OBL}]}\ \text{Voice}_{[\bar{\text{A}},\ \text{OBL}]}\ \dots\ <\text{WH}>\ ]\ ]$

c. Step 3: C^0 Agrees with oblique WH:
$[_{\text{CP2}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ \text{WH}_{[\bar{\text{A}},\ \text{OBL}]}\ \text{Voice}_{[\bar{\text{A}},\text{OBL}]}\ \dots\ <\text{WH}>\ ]\ ]$

=(y)a’ and $\varnothing$, realizing the same feature bundle, although other analytical options are possible.

[16] We thank an anonymous reviewer for prompting us to consider more closely why Dir^0 should serve as a stop-over point for *wh*-movement.

d. Step 4: WH moves to Spec,CP2:
$[_{\text{CP2}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ \text{<WH>}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots\ \text{<WH>}\ ]\ ]$

The remaining steps are shown in (48). Voice^0 in the matrix clause first Agrees with the oblique in Spec,CP2, copying its [OBL] feature (Step 5), and the oblique then moves to Spec,VoiceP1 (Step 6). Finally, C^0 Agrees with the oblique (Step 7), triggering its final movement to Spec,CP1 (Step 8).

(48) Subsequent Steps: cross-clausal movement from full CP complement

a. Step 5: Voice^0 Agrees with oblique WH in Spec,CP2, copies [OBL] feature:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{Voice}_{[\bar{\text{A}},\ \textbf{obl}]} \ldots [_{\text{CP2}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots\ ]\ ]\ ]$

b. Step 6: WH moves to Spec,VoiceP1:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots [_{\text{CP2}}\ \text{<WH>} \ldots\ ]\ ]\ ]$

c. Step 7: C^0 Agrees with oblique WH:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Voice}_{[\bar{\text{A}},\textsc{obl}]} \ldots [_{\text{CP2}}\ \text{<WH>}\ ]\ ]\ ]$

d. Step 8: WH moves to Spec,CP1:
$[_{\text{CP1}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ \text{<WH>}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots [_{\text{CP2}}\ \text{<WH>}\ ]\ ]\ ]$

In (49), we schematize oblique extraction from an aspectless clause complement, as in (31), which we treat as a VoiceP-sized small clause (Elkins et al. 2025, see also Scott 2023). The derivation begins with the embedded Voice^0 Agreeing with the in-situ oblique *wh*-expression and copying its [OBL] feature (Step 1). The *wh*-expression then moves to Spec,VoiceP2 (Step 2). Because the embedded clause lacks a CP layer, the oblique is immediately accessible to the matrix Voice^0, which Agrees with it and copies its [OBL] feature (Step 3). The oblique subsequently moves to Spec,VoiceP1 (Step 4). Finally, matrix C^0 Agrees with the oblique (Step 5), triggering its final movement to Spec,CP1 (Step 6).

(49) Extraction from aspectless clause complement

a. Step 1: Voice^0 Agrees with oblique WH, copies [OBL] feature:
$[_{\text{VoiceP2}}\ \text{Voice}_{[\bar{\text{A}},\ \textbf{obl}]} \ldots \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ ]$

b. Step 2: WH moves to Spec,VoiceP2:
$[_{\text{VoiceP2}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots \text{<WH>}\ ]$

c. Step 3: Voice^0 Agrees with oblique WH in Spec,VoiceP2, copies [OBL] feature:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{Voice}_{[\bar{\text{A}},\ \textbf{obl}]} \ldots [_{\text{VoiceP2}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots\ ]\ ]\ ]$

d. Step 4: WH moves to Spec,VoiceP1:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{WH}_{[\bar{\text{A}},\ \textsc{obl}]}\ \text{Voice}_{[\bar{\text{A}},\ \textsc{obl}]} \ldots [_{\text{VoiceP2}}\ \text{<WH>} \ldots\ ]\ ]\ ]$

e. Step 5: C^0 Agrees with oblique WH:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{WH}_{[\bar{\text{A}},\ \text{\textsc{obl}}]}\ \text{Voice}_{[\bar{\text{A}},\text{\textsc{obl}}]} \ldots\ [_{\text{VoiceP2}}\ <\text{WH}>\ ]\ ]\ ]$

f. Step 6: WH moves to Spec,CP1:
$[_{\text{CP1}}\ \text{WH}_{[\bar{\text{A}},\ \text{\textsc{obl}}]}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ <\text{WH}>\ \text{Voice}_{[\bar{\text{A}},\ \text{\textsc{obl}}]} \ldots\ [_{\text{VoiceP2}}\ <\text{WH}>\ ]\ ]\ ]$

Lastly, we examine oblique extraction from an embedded nonfinite clause as in (34), which we analyze as a VP-sized clause that crucially lacks VoiceP (Scott, 2023). The derivation in (50) begins with matrix Voice^0 Agreeing directly with the oblique *wh*-expression inside the embedded VP and copying its [OBL] feature (Step 1). The oblique then moves to Spec,VoiceP1 (Step 2). Finally, matrix C^0 Agrees with the oblique (Step 3), triggering its final movement to Spec,CP1 (Step 4). Because the nonfinite clause is VP-sized and lacks VoiceP, there is no embedded agreement-bearing Voice^0 through which the oblique can move or which could host *=(y)a'*. The absence of *=(y)a'* in the embedded clause therefore follows directly from its structural size.

(50) Extraction from nonfinite clause complement

a. Step 1: Voice^0 Agrees with oblique WH in Spec,VP2, copies [OBL] feature:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{Voice}_{[\bar{\text{A}},\ \textbf{\textsc{obl}}]} \ldots [_{\text{VP2}}\ \text{WH}_{[\bar{\text{A}},\ \text{\textsc{obl}}]}\ \ldots\ ]\ ]\ ]$

b. Step 2: WH moves to Spec,VoiceP:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{WH}_{[\bar{\text{A}},\ \text{\textsc{obl}}]}\ \text{Voice}_{[\bar{\text{A}},\ \text{\textsc{obl}}]} \ldots\ [_{\text{VP2}}\ <\text{WH}>\ \ldots\ ]\ ]\ ]$

c. Step 3: C^0 Agrees with oblique WH:
$[_{\text{CP1}}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP1}}\ \text{WH}_{[\bar{\text{A}},\ \text{\textsc{obl}}]}\ \text{Voice}_{[\bar{\text{A}},\text{\textsc{obl}}]} \ldots\ [_{\text{VP2}}\ <\text{WH}>\ ]\ ]\ ]$

d. Step 4: WH moves to Spec,CP:
$[_{\text{CP1}}\ \text{WH}_{[\bar{\text{A}},\ \text{\textsc{obl}}]}\ \text{C}_{[\bar{\text{A}}]}\ [_{\text{VoiceP2}}\ <\text{WH}>\ \text{Voice}_{[\bar{\text{A}},\ \text{\textsc{obl}}]} \ldots\ [_{\text{VP2}}\ <\text{WH}>\ ]\ ]\ ]$

Taken together, these derivations extend the analysis of *=(y)a'* to long-distance extraction and derive the observed correlation between its distribution and the structural size of the embedded clause. In full CP complements, the oblique passes through the embedded Voice^0 and the matrix Voice^0, yielding *=(y)a'* in both clauses. In aspectless complements, the embedded VoiceP is sufficient to license an embedded instance of *=(y)a'*, but the absence of an embedded CP means that the oblique proceeds directly from the embedded VoiceP to the matrix VoiceP. In nonfinite complements, by contrast, the embedded clause is only VP-sized and therefore contains no feature-bearing Voice^0; consequently, the oblique cannot trigger *=(y)a'* within the embedded clause and is directly accessible for Agree with matrix Voice^0. The analysis captures the distribution of *=(y)a'* across the three clause types in terms of the availability of the relevant agreement-bearing heads along the movement path, and follows directly from our account of successive-cyclic movement in monoclausal contexts. The SJO data therefore provide evidence that *wh*-movement proceeds through intermediate positions in the verbal domain not only within a single clause, but also across clause boundaries (Chomsky, 1986, 1995, 2001; Rackowski and Richards, 2005; den Dikken, 2009, 2012a,b; van Urk and Richards, 2015).

5 Implications for Mayan and beyond

Having presented our analysis of the Mam facts, and before concluding, this section turns to a closer comparison with extraction morphology in the K'ichean languages, and the analysis developed in Mendes and Ranero (2021) in §5.1 and §5.3, and considers a potential alternate (DP-intervention) analysis in §5.2.

5.1 Mendes and Ranero (2021): stopover only at CP

This section turns to the differences (despite certain surface similarities) between SJO *=(y)a'* and the roughly analogous element *wi* in K'ichean-branch Mayan languages discussed in Mendes and Ranero (2021): K'iche' and Patzún Kaqchikel.[17] These authors, building on Can Pixabaj (2015), observe that the *fronting particle wi* appears immediately following the predicate when certain *low* adjuncts are Ā-extracted; its presence is obligatory in K'iche' (51), and optional in Kaqchikel. For reasons of space, we focus on examples from K'iche', though Patzún Kaqchikel patterns identically in the relevant respects, apart from its optionality (see Mendes and Ranero 2021 for examples).

(51) K'iche' (adapted from Mendes and Ranero, example (2))
Jawi$_i$ x-at-b'ee *(**wi**) iwiir ___$_i$?
where COMPL-B2SG-go FP yesterday
'Where did you go yesterday?'

Despite *wi*'s surface position on the predicate (they analyze it as a *special clitic* (Zwicky, 1977) that postsyntactically relinearizes to immediately follow the predicate), Mendes and Ranero use its distribution to argue *against* there being an intermediate stopoff point in the verbal domain—essentially the opposite conclusion we have drawn for *=(y)a'* in SJO Mam. This claim is motivated by the licensing of *wi* in different long-distance extraction configurations. In clauses featuring an overt complementizer (which the authors take to indicate a full CP embedding), *wi* appears in every clause along the extraction path:

(52) K'iche' (adapted from Mendes and Ranero, example (17))
Jawi x-∅-ki-b'iij *(**wi**) [$_{CP}$ chi k-e-'e *(**wi**)] ?
where COMPL-B3SG-A3SG-say FP COMP INC-B3PL-go FP
'Where did they say that they would go?'

Like Mam, long-distance extraction is also possible from reduced embedded clauses that are not introduced by a complementizer; these are analyzed as structurally smaller Asp(ect)Ps that lack a CP layer. As the data below show, when the embedded clause is an AspP, the fronting particle only occurs in the embedded clause, and crucially not the matrix clause.

(53) K'iche' (adapted from Mendes and Ranero, example (19))
Jas r-uuk' k-∅-a-rayii-j (***wi**) [$_{AspP}$ k-∅-a-tij *(**wi**) le wa] ?
WH A3SG-RN INC-B3SG-A2SG-desire-ACT FP INC-B3SG-A2SG-eat FP DET food
'With what do you desire to eat the food?'

[17] Mendes and Ranero note that a cognate fronting particle occurs in all K'ichean languages, indicating a common etymological source. Given the difference in form, however, it is not clear that SJO *=(y)a'* is cognate with K'ichean *wi*, although establishing their historical relationship requires further investigation. We leave open the possibility that geographical contiguity between Mamean and K'ichean languages may have facilitated local grammatical diffusion.

The empirical generalization that Can Pixabaj (2015) and Mendes and Ranero (2021) arrive at is given in (54). Note that this generalization does not extend to the SJO pattern described in §§3.2–3.4 and summarized in Table 3.

(54) **Fronting Particle Generalization:** In long distance Ā-extraction of low adjuncts from a single embedded clause, the presence of *wi* in the matrix clause is contingent on the presence of an overt complementizer in the embedded clause. (modified from Mendes and Ranero (21))

The analysis put forth in Mendes and Ranero (2021) to account for (54), in simplified terms, is as follows:

1. Intermediate movement proceeds through Spec,CP; C^0 is a phase head (contra den Dikken 2009).
2. Intermediate movement does not proceed through the verbal domain; v^0 is not a phase head (contra, e.g., Chomsky 2001; see Keine 2016; Keine and Zeijlstra 2025).
3. Assuming a copy theory of movement (Chomsky, 1995; Nunes, 2004), the fronting particle *wi* is the spellout of a lower copy of the extracted oblique, realized via the PF operation *Chain Reduction via Substitution.*[18]

Under this analysis, the locative question in (51), which involves local movement only, is derived as follows (simplifying several details of Mendes and Ranero's analysis). First, the locative adjunct $\mathrm{XP}_{\mathrm{Appl}}$ (the feature is assigned by a silent Appl(icative) head) moves (via Internal Merge) from its base position within the verbal domain to Spec,CP (55a). This process creates two copies of $\mathrm{XP}_{\mathrm{Appl}}$ in the syntactic representation and consequently generates a linearization paradox. Such paradoxes are typically resolved through copy deletion (Nunes, 2004); Mendes and Ranero (2021), however, propose that they can also be resolved through *substitution*, with *wi* serving as the relevant vocabulary item. This process, dubbed *Chain Reduction via Substitution*, occurs in (55b). In the post-syntactic module, *wi* relinearizes leftward, encliticizing to the verbal complex (55c).

(55) Chain Reduction via Substitution in local dependencies

a. Internal Merge delivers copy of $\mathrm{XP}_{\mathrm{Appl}}$ in Spec,CP
$[_{\mathrm{CP}}$ $\mathrm{XP}_{\mathrm{Appl}}$ C $[_{\mathrm{AspP}}$ Asp+v+V $[_{v\mathrm{P}}$ … $[_{\mathrm{VP}}$ …] $\mathrm{XP}_{\mathrm{Appl}}$]]]

b. Chain Reduction via Substitution: substitute (lower) $\mathrm{XP}_{\mathrm{Appl}}$ with /=wi/
$[_{\mathrm{CP}}$ $\mathrm{XP}_{\mathrm{Appl}}$ C $[_{\mathrm{AspP}}$ Asp+v+V $[_{v\mathrm{P}}$ … $[_{\mathrm{VP}}$ …] **=wi**]]]

c. Post-syntactic linearization
$[_{\mathrm{CP}}$ $\mathrm{XP}_{\mathrm{Appl}}$ C $[_{\mathrm{AspP}}$ Asp+v+V**=wi** $[_{v\mathrm{P}}$ … $[_{\mathrm{VP}}$ …]]]]

Mendes and Ranero analyze long-distance dependencies involving a full CP complement (such as example (52)) as in (56). In (56a), $\mathrm{XP}_{\mathrm{Appl}}$ moves first to embedded Spec,CP, then to matrix Spec,CP. In (56b) both the base copy and intermediate copy of $\mathrm{XP}_{\mathrm{Appl}}$ are substituted by *wi.* In (56c), both instance of *wi* are displaced leftward: the lowest *wi* linearizes onto the embedded verb complex, and the higher *wi* linearizes onto the matrix verb complex. This captures the empirical generalization that each verb in a long-distance dependency involving a full CP complement is marked with *wi.*

[18]The two languages differ in whether this substitution is obligatory: in K'iche', it is obligatory, and the fronting particle *wi* is consequently obligatory; in Kaqchikel, it is optional, so the lower copy may be realized as either ∅ or *wi.*

(56) Chain Reduction via Substitution in long-distance dependencies (Full CP)

a. Internal Merge delivers copy of XP_{Appl} in each Spec,CP

$[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V ... $[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V ... XP_{Appl}]]]]

b. Chain Reduction via Substitution: substitute (lower copies of) XP_{Appl} with /=wi/

$[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V ... $[_{CP}$ =**wi** C $[_{AspP}$ Asp+*v*+V ... =**wi**]]]]

c. Post-syntactic linearization

$[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V=**wi** ... $[_{CP}$ C $[_{AspP}$ Asp+*v*+V=**wi** ...]]]]

Finally, long-distance dependencies with reduced (AspP-sized) complements, such as example (53) above, are modelled as follows. In (57a) a copy of XP_{Appl} appears in CP, crucially having landed in no intermediate position. Substitution occurs in (57b), and relinearization in (57c): here, *wi* is realized only the lower verbal complex.

(57) Chain Reduction via Substitution in long-distance dependencies (reduced complement)

a. Internal Merge delivers copy of XP_{Appl} in Spec,CP

$[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V ... $[_{AspP}$ Asp+*v*+V ... XP_{Appl}]]]]

b. Chain Reduction via Substitution: substitute (lower) XP_{Appl} with /=wi/

$[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V ... $[_{AspP}$ Asp+*v*+V ... =**wi**]]]]

c. Post-syntactic linearization

$[_{CP}$ XP_{Appl} C $[_{AspP}$ Asp+*v*+V ... $[_{AspP}$ Asp+*v*+V=**wi** ...]]]]

Taken together, then, Mendes and Ranero's analysis successfully accounts for the generalization in (54). In long-distance extraction from a full CP complement, Internal Merge creates two lower copies of the extracted oblique: one in the base position and one in the embedded Spec,CP. Both are subsequently replaced by *wi*, yielding one instance on the embedded predicate and one on the matrix predicate. By contrast, when the embedded clause is a reduced clause lacking a CP layer, the extracted oblique undergoes no intermediate movement. There is therefore only one lower copy available for substitution—the copy in the base position—and so *wi* surfaces only on the embedded predicate.

Against this background, we show that Mendes and Ranero's analysis does not straightforwardly extend to the SJO Mam data. If we adapt their analysis wholesale to SJO (and treat *=(y)a'* as the spellout of lower movement copies), we would make three predictions that are not borne out. First, if oblique movement proceeds directly from its base-generated position to Spec,CP, as in Mendes and Ranero's analysis of K'ichean, we would not expect more than one instance of *=(y)a'* in a local extraction dependency. However, as shown in §3.1, *=(y)a'* may occur multiple times in a single clause. Our analysis developed in §4.2 argues that this pattern reflects two intermediate movement steps, through Spec,VoiceP and Spec,DirP, in addition to the movement from the base position to Spec,VoiceP and ultimately to Spec,CP, as schematized in (58). Under our analysis, the number of intermediate movement steps corresponds directly to the number of (possible) *=(y)a'* exponents. The analysis of K'ichean *wi*, applied to SJO, would not predict this pattern.

(58) [$_{\text{CP}}$ WH C … [$_{\text{DirP}}$ <WH> Dir=***(y)a'*** [$_{\text{VoiceP}}$ <WH> Voice=***(y)a'*** … <WH>]]]

Second, the pattern found in extraction from reduced clauses is different in SJO Mam from the K'ichean pattern. As discussed in §3.3, extraction from an aspectless embedded clause in SJO licenses *=(y)a'* in both the embedded and matrix clauses, even though these aspectless embeddings lack a CP layer. Under Mendes and Ranero's analysis, a reduced clause without CP provides no intermediate position through which the extracted oblique can move. We would therefore expect only the copy in the base position to undergo substitution, yielding a single instance of the extraction marker. The presence of *=(y)a'* in both clauses, therefore, is unexpected on this approach.

Finally, extraction of an oblique from a nonfinite embedded clause does not yield *=(y)a'* within the embedded clause (see §3.4). This is problematic for a Mendes and Ranero-style analysis for SJO. If *=(y)a'* spells out a lower copy of the extracted oblique, we would expect it to occur in the clause containing the base position of the extracted phrase.[19]

We conclude that, while Mendes and Ranero's Chain Reduction via Substitution analysis provides an elegant account of the K'ichean pattern, it cannot be directly extended to SJO Mam. At the same time, we note that our Ā-agreement analysis which requires intermediate stopover points in the verbal (and direction) domain does not straightforwardly extend to the K'ichean pattern: if *wi* in K'ichean were treated as a reflex of extraction through the *v*/Voice domain, our analysis would not capture the fronting-particle generalization in (54) without additional stipulations. These differences suggest that the superficial similarity between the two systems should not be taken to indicate a shared derivational mechanism.

We further conclude that the SJO facts align more closely with those of unrelated languages such as Avatime (Kwa; Ghana; Major and Torrence 2024, Chamorro (Austronesian; Guam and Mariana Islands Chung 1994; Lahne 2009; Georgi 2014), Dinka (Nilotic; South Sudan; van Urk 2015, 2018; van Urk and Richards 2015, Defaka (Ijoid; Nigeria; Bennett et al. 2012), and Sm'algyax (Tsimshianic; Canada and USA; Brown 2024a,b), which likewise exhibit reflexes indicating a close relationship between a functional head in the verbal domain and an extracted element in both local and long-distance dependencies.

5.2 DP intervention?

A recent proposal in Keine and Zeijlstra (2025) raises a further possibility concerning the source of the clause-internal successive cyclicity proposed here. They argue that at least some instances of successive-cyclic movement through clause-medial positions (including in Dinka and Defaka, mentioned in the previous section) need not be attributed to phases or any other independent property of a clause-medial functional head. Instead, they propose that such movement (and its reflexes) can arise from *DP-intervention.* In their analysis, following Aldridge (2004); Coon et al. (2021) and others, an Ā-probe on C^0 may be specified to attract only the *structurally closest* DP (59).

[19] An anonymous reviewer asks whether nonfinite verbs might be subject to a morphophonological restriction that prevents them from hosting *=(y)a'*, thereby providing an alternative explanation for its absence in the relevant contexts. We are not aware of any such restriction. In particular, verbs bearing the nonfinite suffix *-l* may otherwise host enclitics, as shown in the example below.

(i) Ma chin uul=e' k'ayi-**l**=t-e
PROX B1S come=1S see-INF=A2/3S-RN:PAT
'I came to see him' (Pérez Vail 2014:107)

(59) *$\bar{A}$-attraction of the closest DP*
An $\bar{A}$-probe can be specified to only attract the structurally closest DP.

When a DP to be extracted is separated from the probe by an intervening DP, the extracted element must first move to a position above the intervener before it can be attracted to the higher probe. In languages where such "leapfrogging" is possible, this produces an intermediate landing site within the clause and hence an apparent reflex of successive-cyclic movement. The main idea is sketched below, simplifying significantly the details of their analysis. In (60) we see a configuration involving two DPs in a relevant shared domain. If the $\bar{A}$-probe on C^0 is specified to only attract the structurally closest DP, in this case DP$_1$, it cannot straightforwardly Agree with DP$_2$.

(60) ✗ DP$_1$ intervenes between C and DP$_2$
[$_{CP}$ C$_{\bar{A}}$ [$_{XP}$ DP$_1$ … DP$_2$]]

This configuration is ameliorated if DP$_2$ independently moves to a higher structural position than DP$_1$, becoming the structurally closest DP to C^0, (61a). C^0 may now Agree with DP$_2$ (61b), driving movement to Spec,CP (61c). This is sketched below:

(61) ✓ Leapfrogging obviates DP-intervention effect
a. [$_{CP}$ C$_{\bar{A}}$ [$_{XP}$ DP$_2$ DP$_1$ … <DP$_2$>]]
b. [$_{CP}$ C$_{\bar{A}}$ [$_{XP}$ DP$_2$ … DP$_1$]]
c. [$_{CP}$ DP$_2$ C$_{\bar{A}}$ [$_{XP}$ <DP$_2$> … DP$_1$]]

Keine and Zeijlstra (2025) argue that, at least in certain languages, clause-internal extraction reflexes arise via this operation of "leapfrogging" rather than obligatory movement through a clause-medial position. Their proposal is potentially relevant to Mam, as this style of analysis has been invoked to account for the ban on the extraction of ergatives in Mayan (the EEC) (Coon et al., 2021), and Mam is sensitive to the EEC. Additionally, given that *=(y)a'* is restricted to the extraction of obliques, non-core arguments, and adjuncts, this analysis might be fruitful given that such elements, as far as we are aware, entail the presence of at least one core-argument (by hypothesis structurally closer, and therefore a possible intervenor). Could then, the movement path associated with *=(y)a'* be a consequence of leapfrogging?

We tentatively argue that a DP-intervention is *not* ideal for the Mam facts, for the reason that it does not straightforwardly predict the possibility of multiple occurrences of the enclitic within a single clause. In examples such as (62), repeated from (22), there is one one argument DP (a grammatical subject) that plausibly intervenes between the extracted element and the higher probe, and yet *=(y)a'* is realized more than once, on both the verb and the directional auxiliary.

(62) Multiple instances of *=(y)a'* with intransitive predicate
Ja jaw(a') xhchin(a') Luch?

ja	∅-∅-jaw**(=a')**	xhchi-n**(=a')**	Luch
where	COMPL-B2/3S-DIR=**MVMT**	shout-ANTIP=**MVMT**	Pedro

'Where did Pedro shout?'

A more extreme example is given in (63). Here we see a locative extraction configuration with an intransitive predicate, its sole argument, two directionals, and *three* instances of *=(y)a'*: one on the predicate, and one on each directional.

(63) Multiple instances of *=(y)a'* with intransitive predicate
Ja e' b'aj(a') el(a') laqj(a') kwetiy?

ja	∅-e'	b'aj=a'	el=a'	laqj=a'	kwetiy
where	COM-B2/3PL	DIR=**MVMT**	DIR=**MVMT**	explode=**MVMT**	fireworks

'Where did the fireworks go off?'

If each occurrence of the enclitic corresponds to a distinct intermediate position forced by an intervening DP (as we might predict), the presence of multiple reflexes would therefore require further explanation. Thus, while DP-intervention provides a potentially attractive alternative explanation for why oblique extraction might involve a clause-internal intermediate position, the SJO facts suggest that the relationship between intermediate movement and extraction morphology is not reducible to DP-intervention alone.

5.3 Which obliques trigger a reflex?

We conclude our discussion by considering the class of obliques/adjuncts that trigger the movement enclitic when extracted in Mam, and how this differs from the related K'ichean languages. As we've shown above, the extraction of the following obliques optionally triggers the movement enclitic *=(y)a'* on the predicate in SJO: instruments (§2.2.1), benefactives/datives (§2.2.2), locatives (§2.2.3), reasons/purposes (§2.2.4), and manners (§2.2.5). Temporal extraction is unique in that it is *not* associated with the appearance of *=(y)a'* (§2.2.6).

In K'ichean, a distinct subset of obliques triggers the appearance of *wi* under extraction. Mendes and Ranero (2021) identify that it is specifically the *low* obliques which trigger *wi*; clausal or *high* obliques, such as reason and temporal adverbs, do not. Compare, for instance, (64) with (51). The contexts that trigger the movement enclitic in Mam and K'ichean are largely, though not entirely, overlapping.

(64) K'iche' (adapted from Mendes and Ranero, example (14a))

Achike	ru-ma	x-∅-samäj	**(*wi)**	ri	a	Juan?
what	A3SG-RN	COMPL-B3SG-work	FP	DET	CLF	Juan

'Why did Juan work?'

Table 4 summarizes the environments in which the relevant movement enclitic appears in both Mam and K'ichean.[20]

[20] The extraction of manner obliques was not tested by Mendes and Ranero, but Patal Majzul (2007) gives an example of manner oblique extraction from an unspecified dialect of Kaqchikel which does not feature *wi* (p. 80); it is likewise impossible in K'iche', as confirmed by Sapón and Can Pixabaj (2000):204–205.

Oblique Type	**SJO Mam**	**K'ichean**
Instruments	✓	✓
Benefactives/datives	✓	✓
Locatives	✓	✓
Temporals	✗	✗
Reasons/purposes	✓	✗
Manners	✓	✗

Table 4: Comparison of oblique extraction across SJO Mam and K'ichean

Mendes and Ranero encode the difference in behavior of the K'ichean obliques by appealing to a shared structural position in the clause: they take all low obliques to be Merged the specifier of an Appl(icative)P, and as such are endowed with the feature [APPL]. As sketched above in §5.1, the substitution of the movement copy with *wi* is sensitive to this feature. Because high obliques are Merged in a position higher than Spec,ApplP, and therefore lack [APPL], the fact that they do not trigger *wi* is expected. In Mam, however, we do not observe a similar divide into "low" and "high" obliques. In Mam, both reason/purpose obliques and manner obliques are associated with *=(y)a'*, but these do not trigger *wi* in K'ichean (highlighted in the table above).

How can we understand this mismatch? One possibility is that the distinction between high and low obliques varies across languages. On this view, 'why' would be base-generated in a high position in K'ichean but in a lower position in SJO Mam. There is evidence for variation of this kind even within K'ichean. In one dialect of K'iche', for instance, López Ixcoy (1997, 410–413) reports that *wi* is required with extracted indirect objects (datives) but not with benefactives. In another dialect of the same language, Sapón and Can Pixabaj (2000, 204–205) report that *wi* is required with benefactives. A thorough investigation of which obliques pattern as high or low across K'ichean and Mam dialects is beyond the scope of this paper, but further comparative work may shed light on this question.

A related question concerns the fact that temporal obliques do not trigger the movement enclitic when extracted in both Mam and the K'ichean languages, potentially suggesting an important property distinct from other obliques. We currently have no explanation for this restriction. One possibility is that temporal *wh*-items are adverbial rather than nominal and therefore lack (abstract) Case, which we understand to be the feature unifying the relevant obliques that trigger *=(y)a'* when extracted (though see Emonds 1976, 1987; Larson 1985; McCawley 1988; Caponigro and Pearl 2009). However, even complex temporal phrases such as *jni' hor* 'what hour (of the clock)' in (65) fail to trigger the enclitic, despite appearing nominal. We leave the precise featural or categorial characterization of *=(y)a'*-triggering items open, noting that the same issue arises in the other languages discussed here.

(65) *Jni' hor kub' kjone' chib'j?*

jni'	hor	∅-∅-kub'	k-jon(***=a**)=e'	chib'j?
how.many	hour	COMPL-B2/3S-DIR	A2/3PL-cut-DS=MVMT=2PL	meat

'At what time (lit. what hour of the clock) did y'all cut the meat?'

Another possibility, inspired by Mendes and Ranero, is that in both Mam and K'ichean, temporal obliques are Merged high, appearing in the clausal periphery without undergoing *wh*-movement. On this view, the fact that temporal obliques alone are Merged high in SJO could follow from their sensitivity to temporal and aspectual information. A language-specific explanation would nevertheless be needed for why manner and reason/purpose obliques pattern similarly in K'ichean.

These two possibilities—structural height or abstract Case—may be fruitful in understanding another point of difference between the languages: namely that reasons obliques trigger *=(y)a'* when extracted in SJO, but these do not trigger *wi* in K'ichean. In SJO, at least, *tiqu'n* 'why' is plausibly analyzed as a PP, or more specifically as a relational noun phrase, composed of *ti* 'what' and *qu'n* 'because'. The latter is a form of *tu'n*, a relational noun that introduces purposes and reasons (see examples (17) and (18)), as in (66). The PP-like structure of *tiqu'n* is further supported by the existence of other forms corresponding roughly to 'why' (67):

(66) *Jaw xlik'puna qu'n e' jaw chiyon tx'yan*
∅-∅-jaw xlik'pun=a qu'n ∅-e' jaw chiyo-n tx'yan
COMPL-B2/3S-DIR jump=2SG because COM-B2/3PL DIR bark-AP dog
'You jumped because the dogs barked.'

(67) *Alqu'n jaw xlik'pun(a') Xwan?*
al-qu'n ∅-∅-jaw xlik'pun**(=a')** Xwan
who-RN COMPL-B2/3S-DIR jump=MVMT Juan
'Because of whom did Juan jump?'

Thus, reason adjuncts pattern like PPs in SJO. As noted above, however, this is somewhat unexpected, given that both *reason* and *purposive* 'why' pattern identically, despite having been argued to originate in different positions in the clause (Collins 1991; Starke 2001; Buell 2011; Shlonsky and Soare 2011). For example, Starke (2001) argues, based on English, that *reason* 'why' is base-generated in the left periphery, whereas *purposive* (which he calls "motivational") 'why' is base-generated in the VP region. As we have shown, however, both kinds of *why* in SJO license *=(y)a'*. Moreover, SJO lacks a simplex form corresponding exactly to 'why', instead using a PP-based expression, which motivated our analysis of its occurrence with the enclitic. It is therefore notable that in Patzún Kaqchikel, 'why' also appears to be PP-like but does not trigger *wi*, as Mendes and Ranero observe in (68). The contrast between SJO and Kaqchikel therefore remains unexplained.

(68) Achike ru-ma x-Ø-samäj (*wi) ri a Juan?
what A3S-RN COM-B3S-work (*FP) DET CLF Juan
'Why did Juan work?' (Mendes and Ranero, (14a))

6 Conclusion

This paper examined oblique extraction in San Juan Ostuncalco Mam and argued that the movement enclitic *=(y)a'* provides an overt morphological footprint of the path taken by an extracted oblique through the extended verbal (and directional) domain. We implemented this proposal by analyzing *=(y)a'* as the morphological realization of Ā-agreement between the extracted oblique and successive heads (Voice^0 and Dir^0) along the movement path.

We compared SJO *=(y)a'* with the oblique-extraction marker *wi* in the closely related Mayan languages K'iche' and Kaqchikel (Mendes and Ranero, 2021). Although both elements are associated with the extraction of non-core arguments and adjuncts and may occur in the verbal domain, the Mam and K'ichean systems differ in the classes of obliques that trigger their respective reflexes, their distribution in reduced clauses under long-distance extraction, and the possibility of multiple occurrences within a single clause. These differences suggest that superficially similar extraction reflexes need not have the same derivational source, even in closely related languages.

More broadly, if the analysis presented here is correct, SJO provides evidence that at least a subset of obliques pass through one or more lower positions in the clause on their way to Spec,CP. SJO thus contributes to a growing body of work arguing for intermediate positions in the verbal domain in Ā-movement, including work on Avatime (Major and Torrence 2024), Defaka (Bennett et al. 2012), Dinka (van Urk and Richards 2015), and Indonesian (Cole et al. 2008). At the same time, the differences between the SJO and K'ichean systems discussed here suggest that the distribution and interpretation of extraction morphology must be carefully investigated on a language-by-language basis. Further work on extraction-tracking morphology across Mamean and K'ichean languages, particularly Western Mam dialects where cognate movement morphology is also attested (e.g., Tacaná Mam; Munson 1984), will help shed light on the issues raised here.

Abbreviations

Glosses not following the Leipzig Glossing Conventions are as follows: A – "Set A person marker"; ACT - "Active"; B - "Set B person marker"; DET - "Determiner"; DIR - "Directional"; DIST – "distal aspect"; DS - "Directional suffix"; FP - "Fronting particle (K'ichean)"; INC - "Incompletive aspect"; MVMT - "Movement enclitic"; NF - "non-finite" ; OV - "Object Voice (Dinka)" RN - "Relational noun".

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